Supplemental Material to "Explicit masses and the chiral limit in odd reductions of almost-commutative geometries"

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This is the supplement of the paper *Explicit masses and the chiral limit in odd reductions of almost-commutative geometries*. Its parts keep the numbers of the full Supplemental Material, so that a citation of a part in the paper refers to the part of the same name here.

Parts contained. Lemma parts: 17, 27. Other parts: R5, R13, R20 (the letters H, L and R mark the history of the field, the literature check and the referee reports).

The full Supplemental Material, with all parts of the project, exists as `supplemental.pdf` and `supplemental_mobile.pdf` in the same folder as this file. A reference inside a part to a part not listed here points to the full version.

Contents

Part 17 **1**

Part 27 **7**

Part R5 (referee report, main draft and note) **11**

Part R13 (referee report, lemma 27 and additions to lemma 26) **20**

Part R20 (referee report, lemma 17) **24**

Part 17

Explicit masses in the reduced triple, the chiral limit and the set of reductions

Routine part of STRATEGY T10. Proved by Claude Opus 5.5 without Astra or Fable. The identities of Theorems 17.1 and 17.6 were checked with numpy on random finite models (section "Numerical check"). Refereed by Terra 6.1 (gpt-6.1-terra, effort max, 3 Nov 2026; report): of 10 items, 1 correct, 4 needed amendments, and 5 were wrong as first written (Corollary 17.4, Remark 17.7, Proposition 17.8, Proposition 17.9 and the consequence for T10, each by overreach). All corrections are incorporated below. Proposition 17.10 is the referee's.

Outcome. The fermion action of a product M x F, restricted to its physical subspace, is exactly the action of the odd triple M x T on an unconstrained spinor, with the Yukawa block becoming the constant matrix d in D_M + gamma_5 d (Theorem 17.1). A paired Dirac sector gives one Dirac action for either sign s, up to a change of Grassmann variables (Corollary 17.2); other sectors, such as a neutral one, have Majorana-type actions. On a subalgebra commuting with D_F there are no scalar fluctuations, so the mass is explicit (Proposition 17.3). At M = 0 the symmetry of the quark sector is U(N_f) x U(N_f) (Proposition 17.5). The admissible involutions of the quark sector are the Sigma in U(N_f) with Sigma* M Hermitian, which are

exactly the stationary points of the mass term Re tr(M Sigma*); at M = 0 they form a homogeneous space of the chiral group (Theorem 17.6). All of this is kinematic. A pion kinetic term does not come from the fixed spectral action (Proposition 17.10). The condensate requires a measure on gauge fields, infinite volume and an order of limits (Proposition 17.8).

"D = 9" below is the formal KO-label 4 + 5 of Theorem D. The metric dimension of M x T is 4. The label exists only where an involution of sign s = -1 exists: charged paired sectors have one, and the strict light spectrum with one or three unpaired neutrinos has none, since its H_0 has odd dimension (lemma 06, Result 7).

Setting

Let M be a compact Riemannian spin 4-manifold with (D_M, J_M, gamma_5), where J_M^2 = -1, J_M D_M = D_M J_M, J_M gamma_5 = gamma_5 J_M and gamma_5 D_M = -D_M gamma_5. Let F = (A_F, H_F, D_F, J_F, gamma_F) be a finite real triple with the KO-6 signs. On L^2(S) (x) H_F put

```
D = D_M (x) 1 + gamma_5 (x) D_F,      J = J_M (x) J_F,      Gamma = gamma_5 (x) gamma_F,
```

with J^2 = -1, J D = D J and J Gamma = -Gamma J. Let H^+ be the Gamma = +1 subspace. The fermion action is (1/2) <J xi, D xi> on Grassmann variables xi in H^+ (Chamseddine-Connes-Marcolli 2007).

Let U be an admissible involution in the sense of Theorem D: U^2 = 1, U unitary, U gamma_F = -gamma_F U, [U, a] = [U, a°] = [U, D_F] = 0 and U J_F = s J_F U. On H_0 = {gamma_F = +1} put d = U D_F and j = U J_F, restricted to H_0, so that j^2 = s and j d = s d j.

The reduced action

Theorem 17.1. Define R : L^2(S) (x) H_0 -> H^+ by R(phi (x) x) = phi_+ (x) x + phi_- (x) U x, with phi_+- the chiral parts of phi. Then R is unitary, and

```
<J R psi, D R psi'> = <jj psi, DD psi'>,      DD = D_M (x) 1 + gamma_5 (x) d,
```

where jj = J_M (x) j for s = +1 and jj = gamma_5 J_M (x) j for s = -1, with jj^2 = -s and jj DD = s DD jj. Hence the action is (1/2) <jj psi, DD psi> on an unconstrained Grassmann spinor psi. Moreover DD^2 = D_M^2 (x) 1 + 1 (x) d^2, and if f(D^2/Lambda^2) is trace class, Tr f(D^2/Lambda^2) = 2 Tr f(DD^2/Lambda^2).

Proof. R is unitary because U exchanges the two gamma_F-eigenspaces. Let UU = 1 (x) U, which commutes with D and is an involution, so that <J xi, D xi'> = <UU J xi, UU D xi'>.

Put r x = (x + U x)/sqrt(2). Then D_F r = r d, since D_F r x = (D_F x + U D_F x)/sqrt(2) with U D_F x = d x and D_F x = U d x. Also (1 + UU) R/sqrt(2) = 1 (x) r. Since UU commutes with D, this gives UU D R = R DD.

For the real structure, J_F x = U j x and U J_F U = s J_F. As J_M preserves chirality,

```
UU J R(phi (x) x) = J_M phi_+ (x) j x + s J_M phi_- (x) U j x = R((J_M phi_+ + s J_M phi_-) (x) j
↪  x) = R jj (phi (x) x).
```

Therefore <J R psi, D R psi'> = <UU J R psi, UU D R psi'> = <R jj psi, R DD psi'> = <jj psi, DD psi'>. The square follows from gamma_5 D_M = -D_M gamma_5. Finally, D is the direct sum of its restrictions to the eigenspaces of UU, on which it is equivalent to D_M + gamma_5 d and D_M - gamma_5 d, and conjugation by gamma_5 sends the first to minus the second, so their squares are unitarily equivalent. ■

So the doubling is removed by the chirality projection, and the physical space is exactly the Hilbert space of M x T.

Corollary 17.2 (a paired sector gives one Dirac fermion, for both signs). Let H_0 = span(e_L, conj(e_R)) with m > 0 and psi = (psi_1, psi_2). For (d_+, j_+) = (m I, sigma_1 K) and (d_-, j_-) = (diag(m, -m), [[0, 1], [-1, 0]] K) of lemma 06, the action is

```
S_+ = <J_M psi_2, (D_M + gamma_5 m) psi_1>     and     S_- = <gamma_5 J_M psi_2, (D_M + gamma_5 m)
↪  psi_1>
```

respectively, with S_-(psi_1, psi_2) = S_+(psi_1, gamma_5 psi_2). After the charge-conjugation matrix is absorbed, the left argument is an independent Euclidean variable psi-bar, in the Grassmann-bilinear sense of Chamseddine-Connes-Marcolli, section 4. With R_5 = exp(i pi gamma_5/4), R_5 (D_M + gamma_5 m) R_5 = D_M + i m. So the action is that of one Dirac fermion of explicit mass m, whichever sign s is used, up to the displayed invertible change of Grassmann variables.

Proof. The total form is antisymmetric, so on Grassmann variables the two cross terms are equal, which cancels the factor 1/2. The rotation uses R_5 D_M R_5 = D_M. ■

This agrees with Theorem D, item 5: at the level of the classical fermion action, s labels two coordinate descriptions of one bilinear form, and no anomaly or theta shift follows from changing s alone. The rotation R_5 is a formal chiral change of both independent variables; in a quantized theory its Jacobian must be included.

An admissible reduction need not have this paired form. For gamma_F = sigma_3, J_F = sigma_1 K, D_F = m sigma_1 and U = sigma_1 we have s = +1, H_0 is one-dimensional, and the action is the Majorana-type (1/2) <J_M psi, (D_M + gamma_5 m) psi>. The neutral sector of the strict light spectrum has this form with m = 0.

Proposition 17.3 (no Higgs on a commuting subalgebra). Let B in A_F be a unital subalgebra with [D_F, B] = 0, and let U commute with B and B°. The inner fluctuation of D by C^infinity(M) (x) B contains only Clifford one-forms, D_A commutes with UU, and D_A reduces to

```
DD_A = D_M(A) + gamma_5 (x) d,
```

with d constant on M. If [D_F, B] != 0, the same reduction holds, and the fluctuation permits a finite scalar field d_A(x) in place of d.

Proof. For a = f (x) alpha and b = g (x) beta, we have a [D, b] = f [D_M, g] (x) alpha beta + f g gamma_5 (x) alpha [D_F, beta], whose second term vanishes. The term J A J^-1 is a Clifford one-form whose finite coefficients lie in B°. Since U commutes with B and with B°, Theorem 17.1 applies pointwise with D_M(A) in place of D_M. In the other case, U commutes with [D_F, beta], so the fluctuation preserves the reduction. ■

Corollary 17.4 (Yukawa couplings to explicit masses). In the representation of lemma 06, Result 5, with B = {(lambda, Q_lambda, m)} = C + M_3(C), u_L and u_R carry the same left character lambda and the same opposite colour action, d_L and d_R carry the left character conj(lambda) and the same opposite colour action, and e_L and e_R pair as well. So the kept mass blocks commute with B, in the strict projection of lemma 06 and also when only a vectorlike top pair is removed. Hence DD_A = D_M(A) + gamma_5 d, with d the direct sum of the kept mass matrices, and no Higgs field.

Two qualifications apply. First, U(B) contains U(1) x U(3), so before unimodularity or an inherited restriction from the Standard Model gauge group, two independent abelian gauge fields act on the surviving charged fermions; the reading "photon and gluons" needs that restriction. Second, a canonically normalized radial field couples to a fermion of fixed mass m_f with strength m_f/v, which tends to 0 if v tends to infinity at fixed gauge coupling. Under the CCM relation (lemma 11), keeping all Dirac masses fixed while M_W tends to infinity is impossible: some Dirac mass must become unbounded. It need not be the top; top dominance is a further hypothesis. The relation holds in the asymptotic matching regime, and the regime near the cutoff is T5b.

The chiral limit

Consider N_f quark flavours, with q_L, q_R in C^{N_f} (x) C^3, and q-bar_L, q-bar_R = J q_L, J q_R. Let D_F act on (q_L, q_R) by [[0, M], [M*, 0]] (x) 1_3, and by the complex conjugate block on the antiparticles. Let M_3(C) act on the antiparticles on the left and on the particles through its opposite action.

Proposition 17.5 (the symmetry restored at M = 0). Let G(M) be the group of unitaries on this sector commuting with both actions of M_3(C), gamma_F, J_F and D_F. Then

```
G(M) = {(V_L, V_R) in U(N_f) x U(N_f) : V_L M = M V_R},
```

acting by V_L, V_R on q_L, q_R and by the complex conjugates on q-bar_L, q-bar_R. So G(0) = U(N_f) x U(N_f), G(m I) = U(N_f)_V for m != 0, and G(M) = U(1)^{N_f}_V for M positive diagonal with distinct entries. With m_u = m_d = 0 and m_s != 0 in one block, G contains U(2) x U(2). If V is also required to commute with the electric charge, through lambda in B, then each V_L, V_R is diagonal between u and d, and G(0) = U(1)^4 for two flavours.

Proof. The colour identity separates particles from antiparticles, since its left action is zero on particles and the identity on antiparticles. The commutant of the colour matrix algebra is scalar on colour, so V = V' (x) 1_3 on each. The grading separates L and R, and J_F fixes the antiparticle part. Commutation with D_F gives V_L M = M V_R, whose adjoint with unitarity gives the other block equation. ■

The nonabelian flavour transformations are not colour gauge transformations, and at M = 0 the axial part is an ungauged symmetry of the reduced triple. If the central U(1) of U(3) is retained in the gauge group, it contains a common vector phase, which an imposed unimodularity condition can remove. The Lagrangian symmetry U(1)_A is anomalous in the quantum theory, which the classical data do not show.

Theorem 17.6 (the set of reductions). Let B contain only the colour algebra. The admissible involutions of sign s on this sector are exactly the operators, tensored with 1_3,

```
U_{Sigma,s} = [[0, Sigma], [Sigma\*, 0]] on (q_L, q_R),     s [[0, conj(Sigma)], [Sigma^T, 0]] on
↪  (q-bar_L, q-bar_R),
```

with Sigma in U(N_f) and Sigma* M Hermitian. This condition holds if and only if Sigma is a stationary point of V_M(Sigma) = -Re tr(M Sigma*) on U(N_f). The reduced mass operator on H_0 = (q_L, q-bar_R) is

```
d_{Sigma,s} = (Sigma M\* (x) 1_3) (+) (s conj(Sigma\* M) (x) 1_3).
```

The group G(0) acts by Sigma -> V_L Sigma V_R*. Hence:

1. At M = 0 the set is all of U(N_f), for each s, and the action is transitive with stabilizer of Sigma = 1 the diagonal U(N_f): the reductions form the homogeneous space (U(N_f) x U(N_f))/U(N_f)_V.
2. At M = m I with m > 0 it is {Sigma : Sigma^2 = 1, Sigma* = Sigma}, the disjoint union of the Grassmannians of C^{N_f}.
3. For M positive diagonal with distinct entries, it is the 2^{N_f} matrices diag(+-1, …, +-1).
4. For M invertible, the unique Sigma with Sigma* M positive is the polar part of M. It is the unique global minimum of V_M, and it is the unique Sigma for which the first block of d_{Sigma,s} is positive.

With the electric charge in B, the admissible set is the intersection of this set with the commutant of the charge. For two differently charged flavours this keeps diagonal phases at M = 0, and only signs +-1 for positive nonzero diagonal masses.

Proof. An admissible U commutes with both colour actions, so it acts on each of q, q-bar as U' (x) 1_3; it reverses the grading, so it is off-diagonal between L and R; U^2 = 1 with U unitary gives the form [[0, Sigma], [Sigma*, 0]] with Sigma unitary. Then U J_F = s J_F U fixes the antiparticle block. Commutation with D_F is Sigma M* = M Sigma* and Sigma* M = M* Sigma. If H = Sigma* M is Hermitian, then Sigma M* = Sigma H Sigma* is Hermitian, so the two are equivalent. Along Sigma exp(i t X), X Hermitian, the derivative of Re tr(M Sigma*) at t = 0 is Im tr(H X), which vanishes for all Hermitian X exactly when H is Hermitian. The formula for d is direct, using q-bar_R = J q_R. Item 1 is immediate. For item 2 the condition is Sigma* = Sigma. For item 3, H^2 = M* M forces H to be diagonal with entries +- m_i. Item 4 is von Neumann's trace inequality, Re tr(M Sigma*) <= tr |M|, with equality exactly at the polar part, and the first block is Sigma H Sigma*. ■

Remark 17.7 (scope). Item 1 is an equivariant statement: the reductions form a homogeneous space of the chiral group, with the same stabilizer as in the symmetry breaking of QCD. The mass condition has an elementary origin, since it says that H = Sigma* M is Hermitian, and for invertible M the positive choice

is the polar decomposition. Its equality with the stationarity equation of -Re tr(M Sigma*) is exact, and it captures all stationary points, including saddles and maxima. Vacuum alignment (Dashen) asks which stationary point minimizes a physical energy, and the data here contain no such energy; the stationarity calculation is our derivation, not an attribution of this reduction theorem to Dashen.

So the theorem shows the shape of the set of reductions. Identifying a reduction with a condensate direction would need an explicit map to a quantum order parameter, which we do not have. In QCD, the anomalous singlet axial direction also means that the full classical U(N_f) cannot be identified without qualification with the pion manifold.

For invertible M, every stationary point other than the polar part has at least one negative eigenvalue of Sigma* M; for singular M there may be a continuum of minima, as for M = diag(1, 0) and Sigma = diag(1, exp(i theta)). A negative entry in the reduced mass block comes from choosing another reduction of one fixed even triple, and is not a physical mass-sign change at fixed fermion measure. A physical sign change is different: for one Dirac flavour in a background with n_0 zero modes, the determinants at m and -m differ by the factor (-1)^{n_0}, the factor associated with a shift of theta by pi.

Spacetime-dependent Sigma

Proposition 17.10 (referee; no kinetic term from the fixed action). Let U(x) be a pointwise admissible involution on M, with c^mu = [D_M, x^mu] and B_mu = (U d_mu U) restricted to H_0. With R defined pointwise,

```
R^-1 UU D R = D_M + c^mu P_- B_mu + gamma_5 d(x),
```

where P_- = (1 - gamma_5)/2. This operator is generally not self-adjoint. So:

1. If Sigma(x) only changes coordinates for the fixed operator D, the spectral action of D does not depend on Sigma, and conjugating D by a smooth unitary preserves its spectrum. Neither produces a two-derivative term.
2. A kinetic term needs an additional choice of operator. For D_new = D_M + gamma_5 d_Sigma(x) on flat M, with M = m I and Sigma* = Sigma, Sigma^2 = 1, the heat expansion of Tr f(D_new^2/Lambda^2) contains

   ```
   (N_c f(0) m^2/(4 pi^2)) int tr(d_mu Sigma\* d^mu Sigma),
   ```

   tracing over both reduced blocks. The sign is positive when f(0) > 0. At M = 0 this operator has no dependence on Sigma.

Proof. The first display is direct differentiation in Theorem 17.1. Since B_mu is anti-Hermitian and P_- c^mu = c^mu P_+, the derivative term has adjoint c^mu P_+ B_mu. For the coefficient, D_new^2 = -(nabla^2 + E) with E containing -i gamma^mu gamma_5 d_mu d, and tr over spinors of the square of that term is 4 tr(d_mu d d^mu d); then the a_4 coefficient (1/16 pi^2) tr(E^2/2) gives (1/(8 pi^2)) tr_{H_0}(d_mu d d^mu d), and the two blocks of d give 2 N_c m^2 tr(d_mu Sigma d^mu Sigma). ■

So the idea of a pion as a spacetime-dependent reduction, in the form we first proposed, does not produce a kinetic term: the kinetic term needs a nonzero m and a new operator, or a quantum effective action. On the even triple, D_M + gamma_5 S(x) with S built from Phi = m Sigma(x) is self-adjoint for every unitary Sigma. Its kinetic term is the Higgs-type term of an inner fluctuation, available when A_F contains flavour algebras acting on L and R, and then flavour gauge fields accompany it. The coefficient is proportional to N_c m^2, as in constituent-quark estimates of f_pi^2 (quoted from memory); here it is a classical spectral-action coefficient, and m is not derived.

What the spectral data say about the condensate

Proposition 17.8. Let D_M(A) have eigenvalues lambda_n, with n_0 zero modes, on M of volume V, and for m > 0 put

```
c_V(m; A) = (1/V) sum_n m/(lambda_n^2 + m^2) = (1/V) d/dm log |det (D_M(A) + gamma_5 m)|,
```

which is formal on a compact four-manifold and needs ultraviolet regularization, with subtraction of divergent terms including m Lambda^2, m^3 log Lambda and possibly curvature terms.

1. With a subtraction compatible with the pairing +-lambda whose finite remainder is O(m), for example odd and analytic in m, at fixed A on compact M, c_V(m; A) = n_0/(V m) + O(m) as m -> 0. A subtraction merely regular at m = 0 may leave a constant. So at finite volume and fixed background, no nonzero limit exists beyond the zero-mode term.
2. Let rho_m(lambda) = lim_{V -> infinity} <(1/V) sum_n delta(lambda - lambda_n)>*m for a measure <.>*m on gauge fields. Assume that the family is equicontinuous at 0 with rho_m(0) -> rho_0(0), that the tails and the ultraviolet subtraction are controlled uniformly in m, and that the infinite-volume limit may be taken inside the integral. Then lim*{m -> 0} lim*{V -> infinity} = pi rho_0(0). This has the form of the Banks-Casher relation as stated by Giusti and Lüscher (the original, Nucl. Phys. B 169, eqs. (49), (50), has neither the factor pi nor an order of limits with the volume); the sign and flavour normalization of a condensate depend on convention, and the hypotheses here are ours.

Equicontinuity alone is insufficient: rho_m(lambda) = (1 + (1/2) sin(1/m)) exp(-lambda^2) gives no limit.

Proof. Item 1: for a cutoff preserving the pairs +-lambda, the determinant has absolute value m^{n_0} times the product of (lambda^2 + m^2) over lambda > 0, and the nonzero eigenvalues are bounded away from 0. Item 2: m/(lambda^2 + m^2) is pi times the Poisson kernel at scale m. ■

So the spectral triple supplies the operator D_M + gamma_5 m, the density rho_A of each fixed operator, and the continuity of m -> 0 at fixed background, since DD^2 = D_M^2 + m^2. The gauge-averaged density, the regulated determinant or Pfaffian measure, the sequence of volumes and the order of limits need further input. A spectral action may contribute to a proposed gauge weight, but a single compact spectral triple does not specify the functional integral or the thermodynamic limit.

A proof of existence of QCD with fermions would supply a measure. It would not by itself give rho_0(0) > 0. The points to check when it appears are those in STRATEGY T10: whether m_q = 0 is included, and whether infinite volume is reached. With the anti-self-adjoint operator slashed-D = -i D_M and a compatible regulator, det(slashed-D + m) = m^{n_0} times the product over lambda > 0 of (lambda^2 + m^2), which is positive for m > 0; this positivity is part of the method of Vafa and Witten.

The chiral limit on the interval

Proposition 17.9. In the scalar-algebra realization F_0 x I_l of Theorem 08.7, DD = D_M (x) 1 + gamma_5 (x) (D_0 + m Gamma_0), so the interval mass is an explicit mass term with the internal grading. The limit m -> 0 has two distinct routes:

1. *Localization (lemma 09).* With bulk mass -mu and mu l > 1, m = lambda_0 = mu sech(kappa_0 l), of order 2 mu exp(-mu l), which is positive at finite parameters. At fixed mu > 0, l -> infinity gives m -> 0 as an exponentially small residue. The relevant control is the dimensionless suppression: an absolute limit needs a scaling hypothesis, since mu = 2/l gives m -> 0 at fixed mu l, and mu l -> infinity does not ensure m -> 0 if mu grows fast enough. The localization and overlap mechanism is that of domain-wall fermions (Kaplan; Shamir), though this is neither lattice operator.
2. *Boundary Yukawa (lemma 12).* At X = 0 the zero modes are exact (Theorem 12.2), with boundary types (a) and (b). For flavour-degenerate bulk masses, D_0 commutes with U(N_Q) x U(N_U) on the whole tower, and at any l the boundary coupling X preserves exactly the stabilizer {(V_Q, V_U) : V_Q X = X V_U}. So X acts as a spurion located on the wall x = l, and the symmetry is exact at finite l when X = 0. In the towerless limit X = l Y_hat, M_F is proportional to X, and for N_Q = N_U the stabilizer is G(M_F) of Proposition 17.5. At finite l, M_F is not proportional to X: for one flavour with zero bulk masses, the light eigenvalue is arctan(abs(X))/l.

The boundary-coupled operator of route 2 is a different construction from the tensor-product truncation of Theorem 08.7, and an identification of the two at finite l would need a proof.

Proof. Route 1 is Theorem 9.4 and Lemma 9.5, and the scaling examples are direct. Route 2 follows from the invariance of the domain of D_X (lemma 12), Theorem 12.7 and Example 12.5. ■

Caveats

1. **Classical data.** Everything here is at the level of the classical fermion operator. Anomalies, the running of masses and confinement do not appear.
2. **QED.** With the photon present, the axial group reduces as stated in Proposition 17.5, and the set of reductions reduces as stated in Theorem 17.6.
3. **Regime.** Corollary 17.4 inherits the regime of the CCM relation.
4. **Label.** A reduction is a rewriting of one even triple. It gives no parent nine-dimensional geometry, in agreement with Theorem D, and the KO-5 label exists only where an s = -1 involution exists.

Consequence for T10

The reduced action is (1/2) <jj psi, (D_M(A) + gamma_5 d) psi> on an unconstrained spinor, whenever the chosen admissible involution exists. A paired charged sector gives one Dirac action of explicit mass for either sign, up to a change of variables; a neutral unpaired sector gives a Majorana-type action and has only the KO-7 reduction, so the formal label 9 is available only for sectors with the required s = -1 involution. The symmetry restored at m_u = m_d = 0 is U(2) x U(2) on the colour-only algebra, and the reductions there form the homogeneous space U(2), with the mass condition picking out the stationary points of the mass term. The spectral data state the operator, the symmetry, this homogeneous space and the density of each fixed operator. The condensate and the pions as dynamical objects need the QCD measure and a quantum effective action.

Numerical check

Python (numpy, scipy), run on 3 Nov 2026 in a temporary directory, not kept. On a random finite model with KO-4 signs for M (dimension 8) and a quark sector with N_f = 3 and random complex M, for three sign patterns of Sigma and both s: U^2 = 1, U gamma_F = -gamma_F U, [U, D_F] = 0 and U J_F = s J_F U hold to machine precision; R is unitary; the form <J R psi, D R psi'> equals <jj psi, DD psi'> to about 10^-14 and is antisymmetric; j^2 = s and j d = s d j; the formula for d_{Sigma,s} holds, after a first version of it was found wrong by this check; finite differences confirm that V_M is stationary at these Sigma; a generic unitary Sigma gives norm([U, D_F]) of about 10. R_5 (D_M + gamma_5 m) R_5 = D_M + i m holds to machine precision. The spin trace in Proposition 17.10 was evaluated with explicit Dirac matrices (tr of the square equals 4 |a|^2).

Part 27

The axial anomaly on the reduced action, and the two-flavour discrete term

For STRATEGY T10 and T13, following lemma 17 and open question 6 of the steer note. Proved by Claude Opus 5.5, using the index theorem and Witten 1982 as references. Refereed by Terra 6.1 (effort max, 3 Nov 2026; report): two errors of interpretation, in Propositions 27.4 and 27.5, and amendments, all incorporated below.

Outcome.

1. On the reduced operator D_M + gamma_5 d, with masses commuting with the gauge field, the regulated axial trace has an exact form at every cutoff: each flavour block contributes its index, weighted by f(m^2/Lambda^2) (Theorem 27.1). The regulated mass term equals it identically (Proposition 27.3). As Lambda -> infinity this is the anomaly with the coefficient 2 N_c tr(T^3 Q^2) = 1, on the action of Theorem 17.1 (Corollary 27.2).
2. The operator of Proposition 17.10 at M = 0, in independent quark and antiquark variables, is the massless Dirac operator with a pure-gauge connection on one chirality. So for two flavours and a reduction Sigma : S^4 -> SU(2) in the nontrivial class, the spectral action is unchanged, and the Weyl Pfaffian transported along connections from 0 to that pure gauge has the relative sign (-1)^{N_c} (Proposition 27.5). The same sign appears in Witten's anomaly, in the discrete term of the pion sigma model, and in the mod-two index on S^5 of the research state. The sigma field represents a class in the

reduced spin bordism of SU(2) in dimension four, and the anomaly a class in Omega_5^Spin(BSU(2)); these agree through the anomaly comparison and are distinct groups. The corpus S^5 is an internal space, and the S^5 here is the compactified S^4 x R of an auxiliary flow.

Setting

Let M be a closed Riemannian spin 4-manifold. Following Corollary 17.2, a paired sector gives one Dirac fermion: we take the operator

```
DD = D_M(A) + gamma_5 d,   d = diag(m_f) on E = C^{N_c} (x) C^{N_f},   m_f >= 0,
```

on L^2(S) (x) E, with action psi-bar DD psi, and with A the gauge connection of the commuting subalgebra: colour and electromagnetism, so that A is diagonal in the flavour index, with charge Q_f on flavour f. For topologically nontrivial backgrounds, E is replaced by the colour and electromagnetic bundles E_f, with charges normalized so that the flavour representations are globally defined. Let T be a Hermitian matrix on C^{N_f}, diagonal in the same basis, with eigenvalues tau_f. Then d and T commute with A, and Theorem 17.1 gives DD^2 = D_M(A)^2 + d^2.

Let f be a cutoff with f(DD^2/Lambda^2) trace class, for example in the positive Laplace class of lemma 18. Choose gamma_5 = +1 on S^+, and let ind_f = n_{f,+} - n_{f,-} be the index of D_f^+ : S^+ (x) E_f -> S^- (x) E_f.

The rotation psi -> exp(i alpha gamma_5 T) psi, psi-bar -> psi-bar exp(i alpha gamma_5 T) sends DD to D_M + gamma_5 d exp(2 i alpha gamma_5 T), since gamma_5 anticommutes with D_M. So the classical action is invariant at d = 0. Its Jacobian, regulated by the spectrum of DD, is exp(-2 i alpha Tr(gamma_5 T f(DD^2/Lambda^2))), up to the orientation convention.

The axial trace

Theorem 27.1. For every Lambda > 0,

```
Tr(gamma_5 T f(DD^2/Lambda^2)) = sum_f tau_f f(m_f^2/Lambda^2) ind_f.
```

Proof. On block f, DD^2 = D_f^2 + m_f^2, with D_f = D_M(A) on that block. The operator D_f anticommutes with gamma_5, so it maps the eigenspace of D_f^2 with eigenvalue mu > 0 isomorphically between the two chiralities. So these eigenspaces contribute zero to the trace with gamma_5, and only ker D_f remains, with weight f(m_f^2/Lambda^2) and signed dimension ind_f. ■

No expansion in m/Lambda enters. By the index theorem,

```
ind_f = (1/8 pi^2) int tr_f(F wedge F) - (N_c/24) p_1(M),
```

with F Hermitian, and with F = Q_f e F_em + g G on block f, tr_f(F wedge F) = N_c Q_f^2 e^2 F_em wedge F_em + g^2 tr_c(G wedge G), since G is traceless. Reversing the chirality convention reverses the index.

Corollary 27.2 (the anomaly on the reduced action). Let f be continuous at 0, and let Lambda -> infinity at fixed masses. If tr T = 0, with the flavours in the same colour bundle, then

```
2 Tr(gamma_5 T f(DD^2/Lambda^2)) -> f(0) 2 N_c tr(T Q^2) (e^2/8 pi^2) int F_em wedge F_em,
```

with no gluonic and no gravitational term. For two flavours with T = T^3 and Q = diag(2/3, -1/3), the coefficient is 2 N_c tr(T^3 Q^2) = 1, so that the right side is f(0) (e^2/16 pi^2) int F_em,mu nu F-tilde_em^mu nu, with F-tilde^mu nu = (1/2) epsilon^mu nu rho sigma F_rho sigma. A singlet T = 1 on N_f flavours gives in addition the gluonic term f(0) 2 N_f (g^2/8 pi^2) int tr_c(G wedge G) and the gravitational term -f(0) (N_c N_f/12) p_1(M). ■

The normalization f(0) = 1 is the usual convention for a regulator, and it is the value that makes the Jacobian the anomaly. For a smooth cutoff class in which the heat expansion can be used, with E and Omega defined by the Laplace-type square, the local form, with a smooth function chi on M inserted, follows at Lambda -> infinity from the a_4 coefficient, (1/16 pi^2) tr(gamma_5 T (E^2/2 + Omega^2/12)), which gives the

integrands above. The other a_4 terms and the lower coefficients have vanishing gamma_5 spin trace in this problem, and the factor exp(-t d^2) commutes with the heat kernel and changes only higher orders in t. So the algebraic coefficient quoted in the steer note, item 6, now has the regulated trace and measure that it lacked.

Proposition 27.3 (the mass term equals it at every cutoff). Let m_f > 0 for all f. Then

```
Tr(f(DD^2/Lambda^2) d T DD^-1) = Tr(gamma_5 T f(DD^2/Lambda^2)) = sum_f tau_f f(m_f^2/Lambda^2)
↪ ind_f.
```

Proof. DD^-1 = (D_M + gamma_5 d)(D_M^2 + d^2)^-1. The term with D_M has zero trace, by conjugation with gamma_5, since all other factors commute with gamma_5. The remaining term is Tr(f gamma_5 T d^2 (D_M^2 + d^2)^-1). On the nonzero spectrum of D_f it vanishes by the pairing of Theorem 27.1, and on ker D_f the factor m_f^2/m_f^2 is 1. ■

Since {gamma_5 T, D_M + gamma_5 d} = 2 d T, the insertion is psi-bar d T psi in this mass convention, pseudoscalar in a conventional scalar-mass basis. So if the Jacobian and the composite mass insertion are given the same specified regularization, this is the integrated Ward identity in a fixed background at every Lambda, with a common factor f(m_f^2/Lambda^2) on both sides. The trace identity alone constructs neither an independently regulated measure nor a local current, and the local identity has a current-divergence term as well. The surviving zero modes contribute the signed count n_{f,+} - n_{f,-}, whereas the scalar density of Proposition 17.8 sees the unsigned count n_0 = n_{f,+} + n_{f,-}.

Proposition 27.4 (a heavy flavour at fixed cutoff). With the weights of Theorem 27.1, for T^3 on a doublet (t, b) with m_b/Lambda -> 0, the electromagnetic coefficient is

```
2 N_c (1/2) (Q_t^2 f(x) - Q_b^2 f(0)) / f(0),   x = m_t^2/Lambda^2.
```

It equals 1 at x = 0. For the heat cutoff it is (4 exp(-x) - 1)/3, which tends to -N_c Q_b^2 = -1/3 as x -> infinity, with gluonic and gravitational regulated traces proportional to f(x) - f(0). ■

This is a trace in the fixed-cutoff family of lemma 18, where the top is heavy at fixed Lambda. An explicitly massive flavour breaks the axial symmetry, so this calculation alone does not establish a Wess-Zumino term in the low-energy theory. Such a term can be required once the retained symmetry, its background fields or spurions, and the anomaly-matching prescription are specified, as in D'Hoker and Farhi (Nucl. Phys. B 248 (1984) 59 and 77, named in the steer note, and not read here). The weights depend on the chosen regulator, and Proposition 27.3 carries them on both sides of the Ward identity. Making f part of a finite-scale spectral action, as in lemmas 18 and 25, does not by itself identify the regulator of the fermion measure with that action or make these weights observable.

The two-flavour discrete term

Take N_f = 2 and M = 0, with electromagnetism off, so that SU(2)_L x SU(2)_R is a symmetry of the reduced sector (Proposition 17.5). By Theorem 17.6, at M = 0 the admissible involutions are U_Sigma, with Sigma in U(N_f), exchanging the two eigenspaces of gamma_F by Sigma. Let Sigma(x) be a smooth map, and take the pointwise reduction of Proposition 17.10. Its operator at d = 0 is

```
D_M + c^mu P_- B_mu,   B_mu = Sigma d_mu Sigma^*,        (1)
```

up to the convention for which side carries Sigma, in independent quark and antiquark variables. On the full H_0 = (q_L, q-bar_R) the two blocks of (U_Sigma d U_Sigma) are Sigma d Sigma^* and Sigma^T d conj(Sigma). On the chirality selected by P_- this is the massless Dirac operator with the pure-gauge connection B of SU(2), and on the other chirality it has no flavour connection. The operator is generally not self-adjoint. The colour connection is unaffected.

Let Sigma : S^4 -> SU(2) generate pi_4(SU(2)) = Z_2, on the round S^4, with colour background zero, so that D_M has no kernel. Such a Sigma cannot be joined to 1 through maps to SU(2). So let Pf_-(A_t) be the square root of Witten 1982 for one Weyl doublet, transported along the path of connections A_t = t B, with the endpoint fibres compared by the gauge transformation.

Proposition 27.5. For the operator (1), with N_c doublets of the chirality selected by P_-, the transport along A_t gives

```
Pf_-(B)^{N_c} = (-1)^{N_c} Pf_-(0)^{N_c}.
```

The spectral action of the fixed operator D does not depend on Sigma. The fixed classical form is unchanged as a form, while its expression in the fixed reduced variables acquires the connection in (1).

Proof. The display counts N_c Weyl doublets, on which we use the main result of Witten 1982: (det i D-slash)^(1/2) of one doublet has the opposite sign after a gauge transformation in the nontrivial class of pi_4(SU(2)) (Eq. (6), p. 325 there, read on 3 Nov 2026). The last sentences are item 1 of Proposition 17.10. ■

This is a relative sign for the stated Pfaffian transport and endpoint identification. The pointwise R_Sigma of Theorem 17.1 changes fermion coordinates, and pulling the original measure through that change preserves the value of the same integral, because the Jacobian of the measure accompanies the transformed action. So the sign is the comparison between that pulled-back measure and the Pfaffian continued along connections, and for N_c odd they differ between the two classes of reductions. Matching this to a pion field needs the sigma model and its anomaly-matching assumptions. The coupling that results has the sign of the discrete term in the N_f = 2 sigma model: Lee, Ohmori and Tachikawa (arXiv:2009.00033, section 2.5 and Eq. (2.29), read in the LaTeX source) define the coupling as (-1)^{N_c [sigma]}, with [sigma] the class in the reduced spin bordism group of SU(2) in dimension four, which is Z_2, on spin four-manifolds. The original discussion is Witten 1983 (*Global aspects of current algebra*), which we have not read. Proposition 17.10 constructs neither a pion action nor its measure: a kinetic term needs a new operator or a quantum effective action, and Proposition 27.5 supplies only the sign.

Remark 27.6 (the relation with the S^5 index). By Witten 1982, for the flow from A = 0 to A = B, the sign equals (-1) to the number of zero modes of the five-dimensional Dirac operator on S^4 x R. The two ends compactify, with the appropriate gauge trivializations, to the nontrivial SU(2) bundle on S^5, and footnote 2 there describes the invariant zero mode for a special connection. The general protected statement is the parity of the kernel. For isospin j the parity is that of 2 tr T^2 = 2 j (j + 1)(2j + 1)/3, which is odd exactly at j = 2k + 1/2 (LITERATURE, section G). So the discrete term, with N_c colours, has the parity of N_c times the mod-two index of the doublet on the nontrivial SU(2) bundle on S^5. After choices of identification, that bundle is the one in section 6.12 of the research state, and the operator D_{S^5,V} there has the same mod-two index, on the generator of Omega_5^Spin(BSU(2)) = Z_2.

The scope of this identification is the bundle and the index. In the research state the S^5 is an internal space, the base of the chain 7 -> 5. In Proposition 27.5 the S^5 is the compactification of spacetime and the auxiliary flow parameter, with the flavour group of the quarks as structure group. We have not proved a relation between those two spaces, or between their SU(2) groups, and open question 6 of the steer note stays open for that part.

Caveats

- The background is fixed in Theorem 27.1 and Proposition 27.3. A measure on gauge fields and the order of limits are separate, as in Proposition 17.8.
- T, d and A commute by hypothesis. So T^3 and the W are excluded: this is the commuting subalgebra of Proposition 17.3. Off-diagonal d is outside Theorem 27.1, where DD^2 has a term gamma_5 [d, D_M].
- Proposition 27.5 has colour background zero and M = 0. For a nonzero colour background the integral is complex in general, and the statement on the sign is on the class of Sigma through the flow, which we have not developed.
- The sign in Proposition 27.5 is a statement on the fermion measure at one configuration. A sum over Sigma needs a sigma model that is not constructed here.
- The regulated Jacobian by the spectrum of DD is one choice. It is gauge invariant under the commuting group, which is why the anomaly falls entirely on the axial current.

Consequence for T10

The chiral limit now has, on the reduced action: the anomaly with its coefficient, the Ward identity at every cutoff, the top against the cutoff, and the discrete term. Still to be supplied are a measure on gauge fields, the map to a quantum order parameter, and the kinetic term of a pion field, which need a quantum effective action.

Numerical check

By Claude Opus 5.5, in Python (fractions, numpy), 3 Nov 2026, in a temporary directory, not kept. These check the arithmetic and the pairing, and the proofs stand without them.

- 2 N_c tr(T^3 Q^2) = 1 and tr T^3 = 0, in exact fractions. The third-generation coefficient for the heat cutoff at x = 0, 0.1, 1, 5, 50 is 1.0000, 0.8731, 0.1572, -0.3243, -0.3333.
- On T^2 x T^2 with fluxes n_1, n_2 and Landau spectra, truncated to 2000 levels: Tr(gamma_5 exp(-t(D^2 + m^2))) = exp(-t m^2) n_1 n_2 to 15 digits, at (n_1, n_2, t, m) = (1, 1, 0.3, 0.7), (2, 3, 0.05, 1.2), (4, 1, 1, 0).
- tr T^2 half-integer at j = 2k + 1/2 and integer at j = 2k + 3/2 for k up to 20, in exact fractions.

Part R5 (referee report, main draft and note)

Referee report on the main draft and the note

By Terra 6.1 (gpt-6.1-terra, effort max), 3 Nov 2026; prompt in ../lemmas/prompts/terra_check_paper_note.md. Reproduced as delivered, with links relativized. The corrections are incorporated in the paper, the note and lemmas 17 and 18.

The main draft contains useful exact calculations, but it needs substantial revision before it is ready for a Physical Review journal. The most consequential issues are the identification of the formal KO labels with geometric endpoints, the claim that collapse with a fixed light mass requires a tuned bulk mass, and several statements whose scope changes between the theorems, Table I, the abstract, and the discussion.

The note preserves the corrected reduced-mass formula and the sign of j_-. Its main mathematical defect is that the stated condition on the condensate subtraction does not suffice for the claimed $O(m)$ remainder.

The numbered verdicts below count distinct statements, including separate occurrences in Table I and the abstract. We read the LaTeX sources as text and did not compile or edit any file. The underlying lemmas were consulted for the relevant statements and hypotheses; their proofs were not re-refereed in full.

Part 1. Main draft

Chirality at the odd labels The paragraph is at towerless_intervals.tex:184.

1. Classical objection — Correct with amendment. An irreducible spinor representation in odd spacetime dimension has no intrinsic Weyl decomposition: the Clifford volume element is central, so there is no nontrivial chirality operator that anticommutes with every Clifford generator. In the usual smooth, closed, product compactification of an odd-dimensional bulk, the standard fermion reduction therefore has a vectorlike four-dimensional content.

State the compactification and fermion assumptions. Odd-dimensional bulk theories can have chiral modes localized on defects or boundaries, and "no chiral fermions" should not be read as excluding all localized modes. The full text of the cited Witten 1981 article was inaccessible in this review; its precise attribution remains unchecked. The mathematical amendment above follows directly from the Clifford representation.

2. Full chiral algebra — Correct with amendment. For the *specified Standard Model representation* of $\mathbb{C} \oplus \mathbb{H} \oplus M_3(\mathbb{C})$, neither proposed odd factorization exists. Let $e_{\mathbb{H}} = (0, 1_{\mathbb{H}}, 0)$. Its represented range lies in one grading sector and has no counterpart in the other. A grading-reversing U commuting with the algebra would have to identify those two ranges. This is the argument in lemma 06, Result 8.

The conclusion depends on the representation, not on the abstract algebra alone. Name the Standard Model bimodule in the sentence.

3. Label 9 and the absence of needed chiral fermions — Correct with amendment. The charged surviving sectors are vectorlike under the reduced $SU(3)\times U(1)_{\mathrm{em}}$ representation and each admits the $s=-1$ factorization. This establishes a sectorwise formal label 9.

It does not give that factorization for the strict *whole* light spectrum with one or three unpaired neutrinos. A neutral Weyl fermion has a real gauge representation but remains a chiral spacetime fermion; it may be expressed in Majorana form. State explicitly whether the neutral sector is removed, paired, or treated separately.

4. Neutral representation and parity — Correct. The neutral gauge representation is real in the reduced model. If the descended antiunitary j obeys $j^2=-1$, its complex representation space has even dimension. Indeed, writing $j=UK$ gives

$$|\det U|^2=\det(-1)=(-1)^{\dim_{\mathbb{C}} H_0}.$$

For a Majorana block the stronger condition is even multiplicity of **each** singular value, including zero, because j commutes with d^2. Odd total multiplicity is one obstruction.

5. Spectrum at fixed distance from a wall — Error. At fixed $\mu>0$, the normalized left-wall profile tends to

$$\sqrt{2\mu}\,e^{-\mu y}.$$

It is nonzero at every fixed finite distance y from the wall. Thus a region at fixed distance from a wall still sees the chiral wall mode, even though its amplitude is exponentially small when μy is large.

The gapped bulk statement applies when the observation region recedes from **both** walls as $l\to\infty$. In that limit the translation-invariant bulk Dirac operator has spectral threshold μ, while the two half-line limits retain the chiral zero modes.

6. Assignment of the wall limit to label 11 — Gap. The wall-mode limit and the KO-7 factorization are established separately. The text does not construct an identification that assigns KO label 11 to the decompactifying interval. The interval product in Theorem D also has a stated KO-5 factorization. Present the label-to-endpoint connection as an interpretation, or give the missing compatibility and limiting construction.

7. Hořava–Witten comparison — Correct with amendment. Boundary chirality is a fair qualitative comparison. With a common orientation and common identification of the ten-dimensional tangent spaces, the two Hořava–Witten boundary gaugino sectors have the same ten-dimensional chirality. In the interval model here, the two limiting modes of one type-(c) fermion have opposite Γ-chirality. Boundary-induced orientations reverse between the two ends, so the chirality convention must be stated.

The constructions also have different content: Hořava–Witten has an E_8 vector multiplet on each boundary and anomaly cancellation involving the bulk, whereas the interval calculation follows two modes of a bulk Dirac operator. The comparison should identify that common feature and those limits. See Hořava and Witten, hep-th/9603142, §§1–2 and their preceding orbifold paper, §2.

8. Physical point at finite l in an even interior — Gap. The results do not relate l quantitatively to the measured Fermi constant or weak angle, nor do they identify the physical Standard Model point with a member of this interval family. State this as a proposed interpretation. The finite KO-6 structure does accommodate chirality; its identification with a measured finite-l point remains to be shown.

Labels by sector **9. Charged paired sectors — Correct.** In the reduced representation, each charged species and its conjugate form an invariant sector for the represented algebra, its opposite, D, and J. The displayed paired-sector construction admits both signs, including with flavour mixing within a species.

10. One or three unpaired neutrinos — Correct with amendment. With the stated zero Majorana block and reduced algebra, the strict light triple admits $s=+1$ and excludes $s=-1$. Add that the neutral

$s = +1$ involution does **not** preserve particle number. If particle preservation is imposed on the strict unpaired spectrum, neither proposed reduction is available; see lemma 06.

11. Right-handed neutrinos with pure Dirac masses — Correct. Retaining the Dirac partners gives paired singular values in the neutral block, so both signs occur under the stated reduced representation.

12. Majorana masses — Correct with amendment. A Majorana term breaks the particle-number hypothesis used to pair U and QU. Generic nondegenerate neutral Majorana data obstruct $s = -1$, while $s = +1$ remains available under the scalar-character assumptions. State the exact criterion: $s = -1$ exists precisely when every neutral-block singular value has even multiplicity. A nonzero Majorana term alone does not exclude it in several generations.

Weighted sum rule and cutoff limit The claims are at towerless_intervals.tex:202 and towerless_intervals.tex:228; their source is lemma 18, Remark 16.

13. Weighted identity — Correct with amendment. The identity

$$8\kappa M_{W,\mathrm{Taylor}}^2 = \sum_f N_f m_f^2\, \rho(m_f^2/\Lambda^2), \qquad \rho(x) = \frac{2[1-(1+x)e^{-x}]}{x^2},$$

follows from the displayed kernel for the heat cutoff $f(z) = e^{-z}$. Its scope is the specified finite collection of doublets, each massive doublet having one massive Dirac member and one massless member, with the trace weights and mass-compatible doublet decomposition used in Proposition 13 of lemma 18. The mass is the **Taylor ratio** $C_W(0)/C_W'(0)$, not the exact pole mass.

Also define the trace on $\mathbb{R}^4$ through a relative heat trace or action density: the unsubtracted vacuum heat trace has infinite volume.

14. Definition of κ — Correct with amendment. In the three-generation normalization, write

$$\kappa = \frac{P + \sum_i d_i w(x_i)}{6}, \qquad N_i = 2d_i, \qquad w(x) = 3\int_0^1 u^2 e^{-xu}\, du,$$

where P is the particle-trace weight of complete massless doublets and $P + \sum_i d_i = 6$. Thus κ is the charged Taylor kinetic coefficient divided by its all-massless value. For the top alone, $P = 9/2$, $d_t = 3/2$, giving $(3 + w)/4$.

"Normalization of the kinetic term" without these weights does not define κ for the displayed general sum.

15. Small-mass form — Correct with amendment. If **all** $x_i = m_i^2/\Lambda^2 \to 0$, then $\rho(x_i) \to 1$ and $\kappa \to 1$, giving the CCM relation to leading order. This proves the stated relation for the heat-cutoff Taylor calculation. The CCM asymptotic formula also has its own cutoff and trace conventions; the displayed ρ is not a general-cutoff form factor.

16. Five-percent departure — Correct. For the top alone,

$$\frac{M_W^2/m^2}{3/8} = \frac{\rho(x)}{\kappa(x)}.$$

The expansions

$$\rho(x) = 1 - \frac{2}{3}x + \frac{1}{4}x^2 + O(x^3), \qquad w(x) = 1 - \frac{3}{4}x + \frac{3}{10}x^2 + O(x^3)$$

give a departure of about 5% in m/M_W at $x = 0.45^2$.

17. Peak — Correct. The stated peak is supported by the formula. For an analytic uniqueness check, set

$$B = 1 - (1+x)e^{-x}, \qquad D = x^3 + 2 - (x^2 + 2x + 2)e^{-x}.$$

Apart from a positive factor, the derivative has the sign of

$$T(x) = 4 - x^3 + (x^5 + x^4 - 8x - 8)e^{-x} + (x^3 + 4x^2 + 8x + 4)e^{-2x}.$$

The power series of $e^{2x}T(x)$ has positive coefficients at degrees 5, 6 and 7, and negative coefficients at every degree at least 8. Dividing by x^7 therefore gives a strictly decreasing function on $x > 0$, with one zero. The Taylor mass has one positive maximum and falls thereafter.

One small **Python 3** computation solved this derivative equation and evaluated the stated mass formula. It gave $x = 2.02428$, $m/\Lambda = 1.42277$, and $M_W/\Lambda = 0.52422$. The computation tests the location; the sign argument gives uniqueness.

18. Heavy suppression — Correct with amendment. A massive diagonal contribution to the unperturbed heat trace is exponentially suppressed. In a mixed massive–massless contribution, the form factor instead has the algebraic behavior

$$\rho(x) \sim 2/x^2.$$

The corresponding mass-source contribution behaves as $m^2\rho(m^2/\Lambda^2) \sim 2\Lambda^4/m^2$. State which trace contribution is exponentially suppressed so that the sentence does not suggest exponential suppression of the mixed loop.

19. Necessity of Λ growing faster than m_t — Correct with amendment. It is an exact necessity for the **top-only heat-cutoff Taylor family** if the limit requires $m_t/M_W \to \sqrt{8/3}$. To see this, use

$$\rho(x) = 2\int_0^1 ue^{-xu}\,du, \qquad w(x) = 3\int_0^1 u^2e^{-xu}\,du.$$

The function $3 + w - 4\rho$ vanishes at zero and has derivative $\int_0^1 u^2(8-3u)e^{-xu}\,du > 0$. Hence the exact top-only ratio differs from its CCM value for every $x > 0$; it approaches that value only if $x \to 0$.

The necessity is broader than the calculation if stated without the top-only and cutoff hypotheses. For example, retain one complete massless lepton doublet and give one member of each other doublet the common mass m. Then $\kappa = (1+11w)/12$. Near zero, $\rho - \kappa = x/48 + O(x^2) > 0$, while at infinity it tends to $-1/12$. Continuity gives a positive x_* at which the full weighted identity equals the unweighted CCM sum exactly. Scaling $m = \sqrt{x_*}\Lambda$ preserves that equality with a nonzero m/Λ. This is a different mass family.

Generated algebra and the cited remark 20. Bicommutant — Correct with amendment. For a unital complex representation on the finite-dimensional V,

$$C^*(Q(A_0)) = Q(A_0)''$$

by the finite-dimensional bicommutant theorem. State that the commutants are taken in $\mathrm{End}_{\mathbb{C}}(V)$, and retain the unital or nondegenerate-representation hypothesis. The quaternionic real algebra in item 4 needs a separate real-algebra statement.

21. Attribution to Connes and van Suijlekom — Correct. Their introduction says that in many interesting examples the C^*-algebra generated by PAP is a "non-informative full matrix algebra." The draft preserves both the scope and the substance of the observation. Their paper also emphasizes the information retained by the operator system itself. Connes and van Suijlekom, arXiv:2004.14115v2, Introduction.

22. Interval block structure — Correct with amendment. The grading commutes with every compressed function, so it forces the generated algebra to be block diagonal. The explicit interval compressions then show that it fills both blocks, giving $M_2(\mathbb{C}) \oplus M_2(\mathbb{C})$.

The grading alone gives containment in that block algebra; it does not prove equality. The two matrices displayed in the main proof sketch lie in different blocks and commute, so those two examples alone do not generate both full blocks. Add the independent symmetric matrices within each block and the argument separating the two summands; lemma 16, example (b) supplies it.

23. Quaternionic form on S^4 — Correct with amendment. For **real functions**, the compressed operators commute with the antiunitary j, $j^2 = -1$, and generate the real algebra $M_2(\mathbb{H})$. The five compressed coordinate matrices provide the generators. The complex C^*-algebra generated by the same coordinates is $M_4(\mathbb{C})$.

Thus j identifies a quaternionic real form; it is an antiunitary symmetry and is not an element of the ordinary complex-linear commutant in item 1. State the real-versus-complex distinction at towerless_intervals.tex:296, not only earlier in item 4.

Reading the negative results The directions in towerless_intervals.tex:306 are reasonable research directions when kept within their stated scope.

24. Odd-dimensional chirality → boundaries and an even interior — Correct with amendment. Boundaries are one way to obtain localized chiral modes. The claimed even interior is an additional proposal; singularities, defects and other changes of hypotheses are also possible directions.

25. Bare interval limits → bulk mass — Correct with amendment. A negative bulk mass with the stated localization supplies a light pair. The boundary-Yukawa family supplies another route; see check 34.

26. Fixed-compression covariance failure → covariant projection — Correct. Projecting onto a gauge-transformed spectral subspace addresses the demonstrated covariance defect. Further matching of actions remains a separate question.

27. Bounded-fermion collapse → tower-scale fermion or changed action/order — Correct with amendment. This follows for the leading spectral action, specified trace, common profile setting, and tower matching considered in Proposition F. It is a useful direction beyond those assumptions, not a general theorem about every gauge action.

28. Inherited order-zero failure → doubled parent — Correct. The tensor-square parent supplies commuting left and opposite actions and thereby gives a constructive route. The two-sheet and grading inputs remain explicit.

29. Fixed-cutoff behavior → joint cutoff limit — Correct with amendment. The joint limit is a valid route in the top-only family. At fixed cutoff the lemma also contains fully massive families with diverging pole locations, so the direction must retain the mass-family and mass-definition hypotheses.

30. No finite radial vacuum → a potential or constraint — Correct. Adding a specified radial potential or imposing a constraint could alter the radial stationarity problem. Neither follows from the spectral action already analyzed.

31. Commutative whole limits → truncation — Correct with amendment. Compression followed by algebra generation is a constructive way to obtain noncommutative finite data. The commutative-limit statement concerns the specified spectral propinquity and whole scalar algebras; it does not rule out other notions of convergence or other parent algebras.

Table I The table begins at towerless_intervals.tex:46.

Check	Entry and verdict	Amendment
32	Two-point space — **Correct.**	The stated mass and minimal truncation are the content of Theorem A.
33	10 to 11 — **Correct with amendment.**	Say "in the type-(c) family, at fixed $\mu > 0$." The KO-label identification remains the gap in check 6.

Check	Entry and verdict	Amendment
34	10 to 9 — **Error.**	A tuned bulk mass is necessary for the stated one-field type-(c) family, not for every interval collapse. In Theorem C's one-flavour model take zero bulk masses and $X(l) = \tan(ml)$. The light eigenvalue is exactly $\arctan\|X\|/l = m$ for small l, and the higher levels diverge.
35	Yukawa block — **Correct with amendment.**	Replace "D_F is the towerless limit" by "the displayed Dirac Yukawa block of D_F is obtained in the boundary-Yukawa limit." The full finite algebra, Majorana block and real triple are not derived there.
36	9 and 11 — **Correct with amendment.**	Specify the reduced representation and the particle-number caveat in check 10. The full chiral algebra admits neither proposed factorization.
37	Heavy W — **Correct with amendment.**	The weighted formula is for the heat cutoff and Taylor mass with the stated doublets. The top/W limit needs top dominance in the CCM expansion.
38	W against the tower — **Correct with amendment.**	State the leading action, trace, gauge-boundary input and common-profile matching assumptions.
39	Chirality — **Gap.**	The table promotes the proposed assignment of the two odd labels to limits and of the physical point to an even interior into a result. Supply the identification or label it as an interpretation.
40	Finite algebra — **Correct with amendment.**	Distinguish the complex generated algebra from the real quaternionic form, and retain the two sheets, restricted algebra and added grading as inputs.

Abstract 41. Collapse and boundary Yukawa data — Error. The abstract repeats the unrestricted "only along a tuned bulk mass" claim and makes the boundary-Yukawa result sound like the continuation of that same tuned family. The counterexample in check 34 applies. State the two different constructions and their scalings separately.

42. KO labels and weighted mass relation — Correct with amendment. State that the charged and neutral factorizations use the reduced representation, and identify the heat cutoff and Taylor mass for the weighted sum. The complete chiral Standard Model algebra has neither factorization.

43. Whole-triple limit and two-sheet parent — Correct with amendment. "Whole triples converge … only to points" needs restriction to the finite limits of the connected commutative metric triples considered

in Proposition G. The two-sheet construction reproduces the specified algebraic data with the listed inputs; it does not derive a connected parent or the full finite Dirac operator.

Discussion, literature, and one additional main-text statement **44. Masses as momenta combined in quadrature — Error.** The discussion at towerless_intervals.tex:302 applies the real-momentum formula to the localized light mode. For that mode $k = i\kappa_0$, and

$$\lambda_0^2 = \mu^2 - \kappa_0^2.$$

The formula $\lambda_n^2 = \mu^2 + k_n^2$ with real k_n applies to the excited tower.

45. Localization alone determines the texture — Error. At towerless_intervals.tex:149, order-one **entries** of $\hat{Y}$ do not by themselves give the stated individual mass pattern. Let

$$F = G = \mathrm{diag}(\varepsilon, 1), \qquad \hat{Y} = \begin{pmatrix} 1 & 1 \\ 1 & 1+\eta \end{pmatrix}.$$

All entries are order one and the trailing minors are nonzero for $\eta > 0$, but $\det(F\hat{Y}G) = \varepsilon^2\eta$. The small singular value carries the additional arbitrarily small factor η. Require quantitative lower bounds on the relevant minors, as well as the stated profile hierarchy, or present the formula as a generic numerical expectation.

46. Index and open problems — Correct with amendment. The ordinary index is stable under the boundary coupling in the stated family. The equivariant statement is made for the stabilizer of X: symmetry breaking changes the group on which the representation-valued index is read. The discussion should distinguish those statements and retain its explicit qualifications on the parent, grading, dynamics and action.

47. Claims about what was not found — Gap. The statements about the chiral-interval compression, tuned collapse, localization-derived Yukawa block and factorization criterion may be accurate, but the draft does not say which databases, years, synonyms, or citation chains were searched. We cannot infer a comprehensive absence from the text.

The source check here was limited. Queries included `"noncommutative geometry" "interval" "spectral truncation" two point chiral 2026`, a localization/Yukawa query, and an odd/even real-triple factorization query. We opened the relevant sections of the sources cited in checks 7, 21, 48 and 49. Forward and backward citation chains and author recency searches were not completed, and the full texts of Witten 1981 and Banks–Casher 1980 were unavailable. Consequently this review does not establish priority or absence in the literature.

A relevant recent neighboring paper is Baptista, arXiv:2506.09126v2, §§1 and 5: it studies chiral couplings to massive gauge fields in smooth Kaluza–Klein models. Its examples use even-dimensional internal spaces; it does not refute the odd-dimensional Clifford statement. It is relevant to the broader discussion of routes to chiral interactions.

48. Positive attributions checked — Correct. Csáki et al., hep-ph/0310355v1, Eq. (5.6) give $\tan(m_n L) = M_D L$, yielding the one-flavour arctangent spectrum after identifying the boundary parameter. Chamseddine and Connes, arXiv:0706.3688v1, Remark 2.9, give the two Hom-space bimodule and flip real structure cited in the draft. Check 21 establishes the Connes–van Suijlekom attribution.

49. Farsi–Latrémolière theorem — Correct with amendment. The issue is visible in arXiv:2404.00240v3, Theorem 3.4 and Eq. (3.11). The proof's displayed estimate requires control of the vertical operator by the total rescaled operator. A scalar example shows the need for an additional condition: on e_n, take $D_v e_n = n e_n$, $D_h e_n = -n^2 e_n$, and $\epsilon_n = (n + 1/n)^{-1}$. Then $D_{\epsilon_n} e_n = e_n$ although the proposed limit on $\ker D_v$ has spectrum $\{0\}$.

Anticommutation or a suitable relative bound are **sufficient repairs**, rather than the only conceivable repairs. Identify the checked preprint version; the published text was not examined here.

50. Proposition G converse — Error. The statement at towerless_intervals.tex:263 uses the previously arbitrary unitary W_n in an "if and only if." Take the constant triple

$$T_n = T_F = (\mathbb{C}, \mathbb{C}^2, \operatorname{diag}(1,2)),$$

with $P = 1$, and let every W_n swap the two basis vectors. The spectral propinquity is zero, while $\Delta_n = \|\operatorname{diag}(2,1) - \operatorname{diag}(1,2)\| = 1$.

The converse needs **there exist** appropriate light spectral projections and unitaries W_n for which $\Delta_n \to 0$, with g_n chosen as the actual exterior gap or a lower bound tending to infinity. This is the quantifier stated in lemma 15, Theorem 15.7.

Part 2. Note compared with the corrected lemma

We compared chiral_limit_note.tex against lemma 17 and its referee report.

51. Sign of j_- — Correct. With the standard Pauli matrix,

$$i\sigma_2 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

The note's $j_- = i\sigma_2 K$ is exactly the matrix in lemma 17. There is no sign discrepancy.

52. Reduced mass — Correct. In the note's basis $H_0 = (q_L, \bar{q}_R)$,

$$d_{\Sigma,s} = (\Sigma M^* \otimes 1_3) \oplus (s\,\overline{\Sigma^* M} \otimes 1_3).$$

On q_L, UD_F gives ΣM^*. On $\bar{q}_R$, the antiparticle block of U contributes the factor s, yielding $s\overline{\Sigma^* M}$. This agrees with the corrected lemma and report.

53. Reduced action and trace — Correct. The theorem retains the trace-class hypothesis, the sign-dependent real structure, the change of independent Grassmann variables in the paired sector, and the warning that the formal chiral rotation has a quantum Jacobian. The Majorana-type unpaired example prevents the Dirac conclusion from being applied to every sector.

54. Commuting subalgebra — Correct with amendment. The note states the no-scalar-fluctuation proposition and the restriction needed before calling the fields photon and gluons. Add the surviving electron pair to the explicit representation check, and state that the conclusion concerns the kept mass blocks and admissible involution. The first paragraph of the abstract should keep the same restriction.

55. Electric charge and flavour — Correct with amendment. The $U(1)^4$ statement needs **two differently charged flavours**, such as u, d. Two flavours of the same electric charge retain a larger flavour symmetry. The note also omits the corrected lemma's corresponding statement for admissible reductions: with charge included, intersect the colour-only admissible set with the charge commutant **and retain** $\Sigma^* M = (\Sigma^* M)^*$. At zero mass for u, d this leaves diagonal phases; positive nonzero diagonal masses leave signs.

56. Stationarity, reduced mass and polar part — Correct. The condition $\Sigma^* M$ Hermitian is equivalent to stationarity of $-\operatorname{Re}\operatorname{tr}(M\Sigma^*)$. The homogeneous-space statement at $M = 0$, the Hermitian involutions for $M = mI > 0$, and the unique polar minimum for invertible M agree with the corrected lemma. "Sign matrices" for positive distinct masses are to be read in a mass eigenbasis.

57. Vacuum alignment and mass signs — Correct. The note says that the stationary points include saddles and maxima, that no physical alignment energy or map to a condensate has been supplied, and that changing a reduction differs from a physical mass-sign change at fixed measure. These preserve the corrections to Remark 17.7.

58. Spacetime-dependent reduction — Correct with amendment. The displayed reduced operator and the statement that it is generally non-self-adjoint agree with Proposition 17.10. The kinetic term belongs to the separately chosen operator $D_M + \gamma_5 d_\Sigma(x)$, with the coefficient tracing both reduced blocks; it vanishes at $M = 0$. State these normalizations beside the formula and emphasize that this does not construct the massless pion effective action.

59. Fixed-background condensate expansion — Error. "A subtraction regular at $m = 0$" does not alone imply $c_V = n_0/(Vm) + O(m)$. For a finite pair-preserving spectral cutoff, subtracting a nonzero constant C is regular at zero and gives

$$c_V^{\mathrm{reg}}(m; A) = \frac{n_0}{Vm} - C + O(m).$$

Require a compatible subtraction whose finite remainder vanishes at zero with $O(m)$, for example an odd analytic subtraction in m, together with the spectral pairing. This issue is also present in the corresponding wording of the corrected lemma.

60. Banks–Casher form and determinant — Correct with amendment. The note retains the added assumptions $\rho_m(0) \to \rho_0(0)$, equicontinuity, tail and subtraction control, and the order of limits. Add $m > 0$, state the spectral-density and condensate normalization, and specify a regulator compatible with the $\pm\lambda$ pairing. The positive determinant formula is for $-iD_M + m$, as written.

The original Banks–Casher full text was inaccessible here, so the precise hypotheses should remain explicitly those of the note rather than be attributed to that paper.

61. Interval routes — Correct with amendment. For the localized branch state $\mu > 0$, $\mu l > 1$, and define $\kappa_0 > 0$ by

$$\tanh(\kappa_0 l) = \kappa_0/\mu.$$

The asymptotic $m \sim 2\mu e^{-\mu l}$ is for $\mu l \to \infty$, and the absolute limit quoted in the note follows at fixed μ. The boundary-Yukawa operator is a distinct construction from the scalar-algebra tensor-product truncation; the corrected lemma explicitly warns that identifying them at finite l needs proof. The note should retain that warning.

62. Fixed-background continuity and quantum input — Correct with amendment. The operator $D_M + \gamma_5 m$ and its square vary continuously with m. The regularized condensate itself need not have a finite limit at fixed background: a zero mode gives $n_0/(Vm)$. Specify the object whose continuity is asserted. The measure, volume sequence and order of limits remain additional inputs.

63. Provenance — Correct with amendment. The note says the identities were checked with Python on finite models. The recorded numerical checks in lemma 17 cover particular identities and examples, not every claim in the note. Name those checked identities. The earlier report also records four amendments in addition to five erroneous items; "corrected five items" should not be read as the whole scope of that report.

Part 3. Readiness for a Physical Review journal

We recommend substantial revision before submission.

1. **Separate the mathematical results from the endpoint proposal.** The introduction moves from formal KO signs to "10 to 9" and "10 to 11" limits before a map from those signs to the interval geometries or measured weak parameters has been constructed. Give the physical hypothesis, the exact results, and the unresolved identification in that order.
2. **Repair the cross-surface overclaims.** Correct the collapse statement in Table I and the abstract, the localized-mode mass formula in the discussion, the flavour-texture claim, the propinquity converse, and the condensate subtraction. State heat-cutoff and Taylor-mass scope wherever the weighted rule is summarized.
3. **Make the proof sketches show the decisive steps.** Theorem B's sketch omits the index and wall/Weyl arguments; Theorem C invokes concavity and compound matrices without defining the quantities that make the bounds work; Proposition H's displayed matrices do not alone establish the generated algebra. Proposition G needs its existential matching unitary stated in the theorem itself. The supplemental references are useful, but a reader should be able to see why each main-text conclusion follows.

4. **Define the notation needed to read the results.** In particular, define Φ_Q, Φ_U, the threshold quantities, and the symbols $f_{T_p}, g_{T_p}, D_p, K_p, \epsilon_p^{F,G}$ in Theorem C. Define κ by its trace weights, distinguish $M_{W,\mathrm{Taylor}}$ from a pole mass, and state the relative-trace or density convention on $\mathbb{R}^4$.
5. **State the status of the literature comparison.** Give a reproducible, appropriately scoped account of the searches behind the "we did not find" statements. Keep the established interval, domain-wall, truncation and spectral-propinquity ingredients visibly distinct from the proposed combination and from the still-open physical construction.

Amendments and errors

- **Chirality and labels:** scope the odd-dimensional objection, the Standard Model representation, the neutral parity criterion, particle preservation, and the Hořava–Witten comparison as in checks 1–3, 7, 10 and 12. The finite-l physical interpretation and the assignment of label 11 remain gaps.
- **Wall and collapse limits:** replace the fixed-distance bulk claim and the unrestricted tuned-bulk claim as in checks 5, 34 and 41. Keep the type-(c) localization and boundary-Yukawa scalings separate.
- **Spectral action:** define the heat cutoff, Taylor mass, trace weights and κ; restrict the asserted necessity of $m_t/\Lambda \to 0$ to the family proved in check 19.
- **Algebras and limits:** distinguish complex bicommutants from quaternionic real forms, expand the interval generation argument, and insert the existential unitary in Proposition G.
- **Discussion and note:** correct the localized-mode momentum formula, add minor bounds to the texture claim, require an $O(m)$-compatible condensate subtraction, and restore the interval-branch, charge, normalization and provenance qualifications in checks 54–63.

Overall: 63 statements checked: 16 correct, 36 correct with amendment, 4 gaps, and 7 errors.

Part R13 (referee report, lemma 27 and additions to lemma 26)

Referee report on lemma 27 and the additions to lemma 26

By Terra 6.1 (gpt-6.1-terra, effort max), 3 Nov 2026; prompt in ../lemmas/prompts/terra_check_27.md. Reproduced as delivered, with links made relative. The corrections are incorporated in lemma 27 and lemma 26.

The trace identity and the two new boundary calculations reproduce under the stated model assumptions. The material errors are in the interpretation of the heavy-flavour weights, the fermion-frame sign, and the summary's claim that a $2M_W$ top must lie in E.

The numerical checks below used Python 3. SymPy was used for the fermion boundary determinant; NumPy, SciPy and mpmath were used for the neutral-vector roots. The neutral calculation also reproduced the known zero-wall-term formula. A deliberately altered boundary condition failed that control. These computations support the displayed derivations; they do not replace them.

Part A

1. Theorem 27.1 and the index formula — Correct with amendment.
In Theorem 27.1, the nonzero eigenspaces of D_f^2 pair between the two chiralities. Trace class makes the sum of their contributions legitimate, so the stated trace identity holds at every cutoff.

The index and bundle conventions need to be explicit. Let $D_f^+ : S^+ \otimes E_f \to S^- \otimes E_f$, and choose $\gamma_5 = +1$ on S^+. Then $\mathrm{ind}_f = n_{f,+} - n_{f,-}$. With the corresponding orientation and Hermitian-curvature convention, the displayed formula is

$$\mathrm{ind}_f = \frac{1}{8\pi^2}\int_M \mathrm{tr}_{E_f}(F_f \wedge F_f) - \frac{N_c}{24}\langle p_1(TM), [M]\rangle.$$

Reversing the chirality convention reverses the index. The curvature decomposition follows because G is traceless:

$$\operatorname{tr}_{E_f}(F_f^2) = N_c Q_f^2 e^2 F_{\mathrm{em}}^2 + g^2 \operatorname{tr}_c(G^2).$$

Here the products are wedge products. For topologically nontrivial backgrounds, replace the literal trivial bundle $M \times \mathbb{C}^{N_c N_f}$ by the appropriate colour and electromagnetic bundles, with charges normalized so that the flavour representations are globally defined.

2. Corollary 27.2 — Correct with amendment.
The factor 2 in the regulated Jacobian comes from transforming both independent Grassmann variables. With $T^3 = \frac{1}{2}\operatorname{diag}(1,-1)$, $N_c = 3$, and $Q = \operatorname{diag}(2/3,-1/3)$,

$$2N_c \operatorname{tr}(T^3 Q^2) = 1, \qquad F_{\mathrm{em}} \wedge F_{\mathrm{em}} = \tfrac{1}{2} F_{\mu\nu} \widetilde{F}^{\mu\nu}\, d\mathrm{vol}.$$

Thus the electromagnetic coefficient in Corollary 27.2 is correct. The gluonic and gravitational terms cancel for $\operatorname{tr} T = 0$, provided the flavours have the same colour bundle. For $T = 1$, they remain; the integrated gravitational contribution to $2\,\mathrm{Tr}$ is

$$-f(0)\frac{N_c N_f}{12}\langle p_1(TM), [M]\rangle.$$

The limit requires continuity of f at zero. The standard anomaly normalization additionally sets $f(0) = 1$; trace class alone supplies neither condition.

For the local statement, assume a smooth cutoff class for which the heat expansion can be used, and define E, Ω by the Laplace-type square of the Dirac operator. The displayed a_4 terms then give the stated density. The omitted lower-coefficient and other a_4 terms have vanishing γ_5 spin trace in this vector-coupled Dirac problem.

3. Proposition 27.3 — Correct with amendment.
The identity in Proposition 27.3 is correct for $m_f > 0$. Conjugation by γ_5 makes the D_M term odd and hence gives it zero trace; the remaining nonzero eigenspaces cancel by chirality pairing. The surviving zero modes contribute $n_{f,+} - n_{f,-}$. The scalar density in Proposition 17.8 instead involves $n_{f,+} + n_{f,-}$.

This has a Ward-identity interpretation if the Jacobian and the composite mass insertion are given the **same specified regularization**. The trace identity alone does not construct an independently regulated functional measure or a local current. In the present mass convention,

$$\{\gamma_5 T, D_M + \gamma_5 d\} = 2dT,$$

so the mass insertion is $\bar{\psi} dT \psi$; after changing to a conventional scalar-mass basis, the corresponding insertion is pseudoscalar. The local identity also has a current-divergence term.

4. Proposition 27.4 — Error.
The displayed coefficient is correct when $f(0) \neq 0$. For the heat cutoff it is

$$\frac{4e^{-x} - 1}{3},$$

with limits 1 and $-1/3$. The gluonic and gravitational regulated traces are proportional to $f(x) - f(0)$.

The interpretation following Proposition 27.4 goes beyond that calculation. An explicitly massive flavour breaks the axial symmetry, and the trace identity alone does not establish a Wess–Zumino term in its low-energy theory. Such a term can be required after specifying the retained symmetry, its background fields or spurions, and the anomaly-matching prescription.

The weights in this calculation depend on the chosen regulator. Making f part of a finite-scale spectral action does not by itself identify the fermion-measure regulator with that action or make these trace weights observables. The fermion measure and matching calculation must also be specified.

5. Proposition 27.5 — Error.
The chiral pure-gauge form in equation (1) is valid after the paired sector is expressed in independent quark and antiquark variables. More precisely, the operator couples $B = \Sigma\, d\Sigma^\dagger$ to one chirality and is generally non-self-adjoint. On the full $H_0 = (q_L, \bar{q}_R)$, the two blocks of $(U_\Sigma dU_\Sigma)|_{H_0}$ are

$$\Sigma\, d\Sigma^\dagger, \qquad \Sigma^T d\overline{\Sigma}.$$

A nontrivial $\Sigma : S^4 \to SU(2)$ cannot be joined to 1 by a continuous path of such maps. To define the relative Weyl Pfaffian sign, use a path of **connections**, such as $A_t = tB$, with the endpoint fibres compared by the gauge transformation. For one Weyl doublet, Witten's equation (6), p. 325, gives "the opposite sign of the square root"; N_c doublets give $(-1)^{N_c}$.

That is an anomalous relative sign for the chosen Pfaffian transport and endpoint identification. The pointwise R_Σ also changes fermion coordinates. Pulling the original measure through that change preserves the value of the same integral; the measure Jacobian must accompany the transformed action. Accordingly, "the quark integral in that frame changes by $(-1)^{N_c}$" is too strong without specifying the comparison of measures. Likewise, the fixed classical form is unchanged as a geometric form, while its expression in fixed reduced variables acquires the connection in (1).

The quoted two-flavour sigma-model coupling is correctly attributed: Lee, Ohmori and Tachikawa, arXiv:2009.00033v2, §2.5 and equation (2.29) define it as $(-1)^{N_c[\sigma]}$ on spin four-manifolds. They explicitly note that "there is no appropriate five-form" for this case. Matching that term to the quark anomaly requires the sigma-model and its anomaly-matching assumptions; Proposition 17.10 alone constructs neither a pion action nor its measure.

6. Remark 27.6 — Correct with amendment.
For the flow from $A = 0$ to the gauge-equivalent $A = B$, Witten relates the Pfaffian sign to the mod-two index on $S^4 \times \mathbb{R}$. The two ends can be compactified, with the appropriate gauge trivializations, to the nontrivial $SU(2)$ bundle on S^5. Witten's footnote 2 describes an invariant zero mode for a special connection; the general protected statement is the **parity** of the kernel.

The nontrivial bundle is, after choices of identification, the bundle $SU(3) \to S^5$ used in research-state §6.12. Its twisted Dirac operator has the same mod-two index. For isospin j, the parity is that of $2j(j+1)(2j+1)/3$, odd for $j = 2k + \frac{1}{2}$.

The outcome paragraph should distinguish the bordism groups: the sigma field represents a class in $\widetilde{\Omega}_4^{\mathrm{Spin}}(SU(2))$, whereas the five-dimensional anomaly is represented in $\Omega_5^{\mathrm{Spin}}(BSU(2))$. Their signs agree in this construction through the anomaly comparison; they are not literally one element of one group. The abstract bundle identification also does not identify the internal gauge group or dynamics with the flavour construction.

Part B

7. Proposition 26.3c — Correct with amendment.
The boundary equation in Proposition 26.3c follows from a self-adjoint bulk-plus-wall problem. For example, normalize $u = (S + A')/\sqrt{2}$, $v = (S - A')/\sqrt{2}$, and take

$$H(f, g, \chi) = (-g',\, f',\, w g_u(\ell)), \qquad f_u(\ell) = w\chi,$$

along with $g_{O'}(\ell) = 0$, $f_v(\ell) = 0$, and the stated conditions at zero. The boundary form cancels. Eliminating χ for $m \neq 0$ gives $f_u = (w^2/m) g_u$.

Let $a = m\ell$, $s = \sin 2\theta$, and $\Delta = \sin^2 a - s^2$. Direct elimination gives

$$2a\Delta \sin a = \omega \cos a\,(\Delta + \sin^2 a).$$

The two nonzero-mode limits reproduce $\Delta \sin a = 0$ and the first referee's $\cos a(2\sin^2 a - s^2) = 0$. At $w = 0$, the decoupled wall Weyl zero mode must also be counted.

At $\theta = \pi/4$, the equation factorizes as stated. The written proof divides by $\cos 2\theta$, so it misses this angle; use the determinant before division. For every $0 < \theta < \pi/2$, put $u = M_W \ell = \arcsin(s/\sqrt{2})$. On $0 < a < u$,

$$\omega = 2a \tan a \, \frac{s^2 - \sin^2 a}{s^2 - 2\sin^2 a}$$

is continuous and strictly increasing from 0 to $+\infty$. This proves the stated light root, including at $\theta = \pi/4$.

For the **displayed** $(10, 6)$ charge-2/3 channel, a state lies below M_W for every finite positive wall mass; in the infinite-mass limit the lowest root of this isolated channel is M_W. The same construction works for the specified $(5, 15)$ plane with chiralities exchanged and the corresponding wall Weyl field. A complete gauge-invariant wall multiplet, other boundary mixings, bulk masses and the other towers require separate analysis. Their presence can add still lighter states, so equality for the lightest state of a completed model is unproved.

8. Result 26.8 — Correct with amendment.
The numbers in Result 26.8 reproduce for the stated tree-level model. At the connector wall, R in the hypercharge combination must mean the **rotated wall direction**

$$L_\ell = (S + A')/\sqrt{2}, \qquad R_\ell = (S - A')/\sqrt{2}.$$

With $p = \cos\phi$, $q = \sin\phi$, the wall conditions are

$$O' = 0, \quad -qR_\ell + pB = 0, \quad \partial_x L_\ell = 0, \quad \partial_x(pR_\ell + qB) = rm^2(pR_\ell + qB).$$

At $x = 0$, S, A, B are Neumann and O is Dirichlet. Eliminating these four amplitudes gives, for the lowest neutral root $a = M_Z \ell$, with $t = r/\ell$,

$$(1 + t\,a\cot a)\sin^2 a = \frac{\sin^2(2\theta)}{2}(1 + p^2 + t\,a\cot a).$$

The definitions of p^2 and r/ℓ in the result follow from the stated zero-angle coupling prescription. Using its inputs, the two recomputed rows are:

θ	$M_Z\ell$	r/ℓ	M_W^2/M_Z^2	4D reference
$\pi/4$	0.990308	1.6336	0.62898	0.76230
1.13774393	0.675757	1.5957	0.70868	0.76067

The quoted shifts, approximately -0.13 and -0.05, therefore reproduce. The four-dimensional number is a meaningful **reference within this matching prescription**. A physical precision comparison needs the model's coupling definitions, wavefunction overlaps, threshold matching and radiative corrections: those can affect the comparison at the same order as a small finite-interval shift. The "about 1.2 TeV" threshold should remain a numerical statement for this prescription, with its parameter range stated.

9. Revised outcome and two regimes — Error.
The claim in the outcome that a top at $2M_W$ must remain in E does not follow from Proposition 26.3c. The

proposition proves a lighter charge-2/3 partner in its displayed channel. It does not exclude identifying a higher eigenstate with the top.

Indeed, let $u = M_W \ell < \pi/4$. Equation (3) has an eigenvalue $m = 2M_W$ when

$$\omega = \frac{4u \sin 2u}{1 + 2\cos 2u}.$$

For example, Python gives $\theta = 1$, $\omega = 2.04350$, a state at $2M_W$, and another at $0.84972M_W$. The higher state has nonzero O_L and R_R components. Its physical identification and the lighter state's couplings still need analysis. Replace the summary by the proved lighter-partner statement.

The low-tower potential, the wall-mass spectrum, the wall kinetic term and the step-threshold running also use different simplified spectra. A combined model needs the altered complete-multiplet determinants and thresholds before its vacuum and matching scale can be asserted together. The qualifications in "The top against the matching scale" correctly treat the Yukawa table as an illustration.

The first referee's requirements do **not** appear as a consolidated list in lemma 26. The first report's amendment list is linked, and several corrections are scattered through the lemma. The opening claim that all are incorporated should distinguish textual corrections from work still needed, including complete wall multiplets, their potential and thresholds, and the finite-angle coupling matrix.

Amendments and errors

- Specify the bundles, chirality convention and cutoff regularity for Theorem 27.1 and Corollary 27.2.
- State the common-regularization assumption behind the integrated Ward identity.
- Correct the Wess–Zumino and regulator interpretation in Proposition 27.4.
- Replace the frame-independent sign claim in Proposition 27.5 by the Pfaffian transport statement and include the measure Jacobian under R_Σ.
- Distinguish the four- and five-dimensional bordism groups and the special S^5 connection.
- Supply the self-adjoint wall-field domain, count the decoupled zero mode, and remove the division by $\cos 2\theta$.
- State the neutral wall boundary conditions and the matching convention behind the four-dimensional reference.
- Replace the claim that a $2M_W$ top must lie in E by the proved lighter-partner result, and separate completed referee corrections from outstanding calculations.

Overall: 0 Correct; 6 Correct with amendment; 0 Gap; 3 Error.

Part R20 (referee report, lemma 17)

Referee report on lemma 17

By Terra 6.1 (gpt-6.1-terra, effort max), 3 Nov 2026; prompt in ../lemmas/prompts/terra_check_17.md. Reproduced as delivered, with links relativized and one link to an unofficial copy of a paper replaced by the journal DOI. The corrections are incorporated in lemma 17. Claude checked by hand the coefficient in Part 2(iii).

Theorem 17.1

Verdict: Correct with amendment. The identities hold for both signs. The trace statement needs a hypothesis ensuring that the traces exist, for example that $f(D^2/\Lambda^2)$ is trace class.

Write $P_\pm = (1 \pm \gamma_5)/2$ and $\widetilde{U} = 1 \otimes U$. Since U is unitary and exchanges the two eigenspaces of γ_F, R is an isometry onto

$$(S^+ \otimes H_F^+) \oplus (S^- \otimes H_F^-) = \mathcal{H}^+.$$

In particular, it is surjective.

Using $[U, D_F] = 0$, $J_M\gamma_5 = \gamma_5 J_M$, and $UJ_F = sJ_FU$, direct calculation gives

$$\widetilde{U}DR = R\mathcal{D}, \qquad \widetilde{U}JR = R\mathcal{J},$$

where

$$\mathcal{D} = D_M \otimes 1 + \gamma_5 \otimes d, \qquad \mathcal{J} = \begin{cases} J_M \otimes j, & s = +1, \\ \gamma_5 J_M \otimes j, & s = -1. \end{cases}$$

The second identity includes the sign on the negative-spin-chirality component. These identities prove the displayed bilinear-form equality.

The claimed real structures have the expected signs:

$$\mathcal{J}^2 = -s, \qquad \mathcal{J}\mathcal{D} = s\,\mathcal{D}\mathcal{J}.$$

The square of $\mathcal{D}$ has no cross term because d is constant and $\{D_M, \gamma_5\} = 0$.

For the trace, the two $\widetilde{U}$-eigenspaces carry operators equivalent to $\mathcal{D}_\pm = D_M \pm \gamma_5 d$. Since $\gamma_5\mathcal{D}_+\gamma_5 = -\mathcal{D}_-$, their squares are unitarily equivalent. This gives the factor 2, subject to the trace-class qualification.

Corollary 17.2

Verdict: Correct with amendment. The two displayed forms give the same Dirac action after an identification of the Grassmann variables. They are not literally the same formula in the same coordinates.

The bilinear form $\langle \mathcal{J}\psi, \mathcal{D}\psi' \rangle$ is antisymmetric for both signs: $\mathcal{J}^2 = -s$ and $\mathcal{J}\mathcal{D} = s\mathcal{D}\mathcal{J}$. On Grassmann variables, exchanging the two cross terms introduces a second minus sign. They therefore agree, and their sum cancels the prefactor $1/2$.

The two components ψ_1, ψ_2 are independent Grassmann generators. After the charge-conjugation matrix is absorbed into a row variable, the first argument defines an independent Euclidean $\bar{\psi}$. This is a Grassmann-bilinear construction; it does not impose ordinary complex conjugacy between integration variables. It agrees with the Grassmann construction in CCM, §4, equations (4.3)–(4.7).

For the two choices in the corollary,

$$S_-(\psi_1, \psi_2) = S_+(\psi_1, \gamma_5\psi_2).$$

Thus the equality is an equality under an invertible change of variables. Also,

$$R_5(D_M + \gamma_5 m)R_5 = D_M + im.$$

This is a formal chiral change of both independent fermion variables. Its quantum Jacobian must be included; the identity alone does not identify the quantum measures.

Proposition 17.3

Verdict: Correct with amendment. The fluctuation and reduction are correct. The statement about JAJ^{-1} should distinguish its Clifford and finite factors.

For a one-form $A = \sum_i a_i[D, b_i]$, the displayed formula follows directly from the product Dirac operator. If $[D_F, \mathcal{B}] = 0$, the finite scalar part vanishes. The remaining terms are Clifford one-forms with coefficients in the represented algebra. In JAJ^{-1}, the **finite coefficients** lie in the opposite representation; the whole operator also contains a Clifford one-form.

Because U commutes with D_F and $\mathcal{B}$, it commutes with $[D_F, b]$ and hence with A. The relation $UJ_F = sJ_FU$ then implies $[U, JAJ^{-1}] = 0$. Thus $[D_A, \widetilde{U}] = 0$, and the proof of Theorem 17.1 applies.

When $[D_F, \mathcal{B}] \neq 0$, the same argument permits a finite scalar fluctuation $d_A(x)$. It may vanish for a particular one-form A; noncommutation permits a scalar field and does not force every fluctuation to contain one. The usual self-adjointness condition on the fluctuating one-form is understood.

Corollary 17.4

Verdict: Error. The bimodule and fixed-mass Yukawa statements are correct, but the asserted heavy-top conclusion does not follow, and "photon, gluons" needs an additional gauge-group restriction.

In the representation of lemma 06, Result 5, restriction to $\mathcal{B} = \{(\lambda, Q_\lambda, m)\}$ gives the same left action on u_L and u_R, namely λ, and the same opposite colour action on both. Likewise, d_L, d_R both carry the left character $\bar{\lambda}$ and the same opposite colour action. The surviving electron pair also has identical left and opposite actions. Hence the kept mass blocks commute with $\mathcal{B}$, including when a vectorlike top pair alone is removed.

For a canonically normalized radial Higgs field, the coupling to a fermion of fixed mass m_f is m_f/v. It tends to zero if $v \to \infty$. But, at fixed nonzero gauge coupling, the matching relation in lemma 11 is

$$8M_W^2 = \sum_{\text{generations}} (m_\nu^2 + m_e^2 + 3m_u^2 + 3m_d^2).$$

Consequently, keeping **all** Dirac masses fixed while $M_W \to \infty$ is incompatible with that relation. If only a specified light mass is kept fixed, some Dirac mass must become unbounded, but it need not be the top mass. For example, the relation permits an unbounded charged-lepton Dirac mass with zero top mass. Top dominance is an additional hypothesis. The relation also belongs to the stated asymptotic matching regime, not automatically to the full finite-cutoff action.

Finally, the unrestricted unitary group of $\mathcal{B}$ contains $U(1) \times U(3)$. Before unimodularity or an inherited restriction from the Standard Model gauge group, two independent abelian gauge fields can act on the surviving charged fermions. For example, the central colour phase acts on quarks and not leptons. The phrase "photon, gluons" therefore requires specifying the residual gauge group and its unimodularity condition.

Proposition 17.5

Verdict: Correct.

In the specified Standard Model representation, the left colour action is zero on particles and defining on antiparticles; the opposite colour action is defining, as an opposite representation, on particles and zero on antiparticles. The represented colour identity therefore separates the particle and antiparticle sectors. The commutant of the full colour matrix algebra is scalar on colour.

Commutation with the grading then gives independent flavour unitaries V_L, V_R, and commutation with J_F fixes their complex conjugates on antiparticles. The remaining condition is

$$V_L M = M V_R.$$

Its adjoint, together with unitarity, gives the other mass-block equation. The listed groups follow. For example, with two zero masses and a third nonzero mass, the group contains $U(2)_L \times U(2)_R$; this statement is for the colour-only algebra.

In the u, d sector, the λ-representation distinguishes the two electric charges. Requiring flavour transformations to commute with it makes each V_L, V_R diagonal, giving $U(1)^4$ at $M = 0$. The physical electromagnetic subgroup also requires the gauge-group normalization discussed under Corollary 17.4.

The following unnumbered sentence, saying that "these groups are not inner," should be read with care. The nonabelian flavour transformations are not colour gauge transformations. If the central $U(1) \subset U(3)$ is retained in the lifted gauge group, however, its action contains a common vector phase; an imposed unimodularity condition can remove that continuous subgroup.

Theorem 17.6

Verdict: Correct with amendment. The classification, signs, critical-point equation and four numbered items are correct. The final charge sentence must retain the mass-compatibility condition, and the colour identity factors should be displayed or explicitly suppressed.

Commutation with both colour actions separates particles and antiparticles and makes every block scalar on colour. Grading reversal and unitary involutivity give the stated particle block. From $UJ_F = sJ_FU$, the antiparticle block is s **times** its complex conjugate, as printed.

On particles, commutation with D_F is equivalent to

$$\Sigma M^\dagger = M\Sigma^\dagger, \qquad \Sigma^\dagger M = M^\dagger \Sigma.$$

The equations are equivalent: if $H = \Sigma^\dagger M$ is Hermitian, then $\Sigma M^\dagger = \Sigma H \Sigma^\dagger$. Conversely, conjugating the first equation by $\Sigma^\dagger, \Sigma$ gives the second.

For a Hermitian X, differentiation along Σe^{itX} gives

$$\frac{d}{dt}\Big|_0 \operatorname{Re}\operatorname{tr}(Me^{-itX}\Sigma^\dagger) = \operatorname{Im}\operatorname{tr}(HX).$$

This vanishes for every Hermitian X exactly when H is Hermitian. With colour shown, the reduced mass is

$$d_{\Sigma,s} = (\Sigma M^\dagger \otimes 1_3) \ \oplus \ (s\,\overline{\Sigma^\dagger M} \otimes 1_{\bar{3}}).$$

At $M = mI$, the condition is $\Sigma^\dagger = \Sigma$, so the components are the Grassmannians indexed by the number of negative eigenvalues. If M is positive diagonal with distinct entries, $H^2 = M^\dagger M$ forces H to be diagonal with entries $\pm m_i$, giving 2^{N_f} choices. For invertible M, $H > 0$ gives the unique polar unitary and the unique maximum of $\operatorname{Re}\operatorname{tr}(M\Sigma^\dagger)$. The first block of $d_{\Sigma,s}$ is $\Sigma H \Sigma^\dagger$, so it is positive in exactly that case.

With electric charge included, the admissible set is the **intersection** of this set with the charge commutant. For two differently charged flavours the latter consists of diagonal phases, but the mass condition remains. For instance, positive nonzero diagonal masses leave signs ± 1, not arbitrary phases.

Remark 17.7

Verdict: Error. The coset and stationarity comparisons are valid with the interpretation in Part 2(i), but the statement about all other critical points and mass signs is too broad.

For singular M, other critical points need not introduce a negative mass eigenvalue. Take

$$M = \operatorname{diag}(1, 0), \qquad \Sigma = \operatorname{diag}(1, e^{i\theta}).$$

Every θ gives $\Sigma^\dagger M = \operatorname{diag}(1, 0)$; all are global minima, and none introduces a negative eigenvalue. For invertible M, every critical point other than the positive polar choice has at least one negative eigenvalue of $\Sigma^\dagger M$. Its numerical signature $n_+ - n_-$ need not be non-positive.

Changing Σ changes the chosen reduction of a fixed even triple. A negative entry in the reduced mass block is therefore not, by itself, a physical mass-sign change at fixed fermion measure. A theta-shift statement requires the associated change of variables and its Jacobian.

The commutator with D_M in the last paragraph is nonzero for varying Σ. Its consequences are worked out in Part 2(iii); a dynamical pion field does not follow from that commutator alone.

Proposition 17.8

Verdict: Error. The formal logarithmic-derivative identity and the fixed-background conclusion are correct with a compatible regularization. The hypothesis in item 2 is insufficient for its limit.

For a finite spectral cutoff preserving the $\lambda, -\lambda$ pairs,

$$|\det(D_M(A) + \gamma_5 m)| = m^{n_0} \prod_{\lambda > 0} (\lambda^2 + m^2).$$

Differentiation gives the stated sum, counting both signs of every nonzero eigenvalue. On a compact four-manifold the sum requires ultraviolet regularization and subtraction. In general the divergent terms include

both an $m\Lambda^2$ term and an $m^3 \log \Lambda$ term, together with possible curvature-dependent terms. With a subtraction scheme regular at $m = 0$, the positive spectral gap away from zero gives

$$c_V(m; A) = \frac{n_0}{Vm} + O(m).$$

Uniform continuity of $\rho_m(\lambda)$ at $\lambda = 0$ does not imply that $\rho_m(0)$ has a limit as $m \to 0$. For example,

$$\rho_m(\lambda) = \left(1 + \tfrac{1}{2}\sin(1/m)\right) e^{-\lambda^2}$$

is nonnegative and uniformly continuous at zero, but its Poisson-kernel integral has no limit as $m \downarrow 0$. A sufficient amendment includes $\rho_m(0) \to \rho_0(0)$, control of the spectral-density tails and ultraviolet subtraction uniformly in m, and justification for taking the infinite-volume limit inside the integral. Under those conditions the Poisson-kernel argument gives $\pi\rho_0(0)$. The sign and any flavour normalization of a condensate depend on its convention.

A fixed operator $D_M(A)$ determines its own spectral measure. The gauge-averaged density, a regulated fermion determinant or Pfaffian measure, the sequence of volumes, and the order and existence of limits require further input. A spectral action may contribute to a *proposed* gauge weight, but an individual compact spectral triple does not by itself specify the required functional integral and thermodynamic limit.

The later assertion that $\det(D_M + im)$ is positive is also convention-dependent and is false literally for a finite paired cutoff: a single $\pm\lambda$ pair contributes $-(\lambda^2 + m^2)$. The standard positive expression uses the anti-self-adjoint operator $\not{D} = -iD_M$:

$$\det(\not{D} + m) = m^{n_0} \prod_{\lambda > 0} (\lambda^2 + m^2) > 0 \quad (m > 0),$$

with a compatible regulator. The factor relating this determinant to $\det(D_M + im)$ carries a phase convention.

Proposition 17.9

Verdict: Error. The stated interval mass and boundary stabilizer are correct under their respective hypotheses, but both routes need qualifications.

The first displayed reduced operator follows from Theorem 08.7 for its **scalar-algebra minimal truncation**, with compatible real structure and interval pair. In the localized regime of Theorem 9.4, $M_{\text{bulk}} = -\mu$ and $\mu\ell > 1$,

$$m = \mu \operatorname{sech}(\kappa_0 \ell) \sim 2\mu e^{-\mu\ell}.$$

It is positive at finite parameters. At fixed $\mu > 0$, $\ell \to \infty$ gives the stated chiral limit.

Without fixing μ, the claim that the limit is reached **only** when $\mu\ell \to \infty$ is false. Let $\ell \to \infty$ and $\mu = 2/\ell$. Then $\mu\ell = 2$, while m is a fixed positive constant times $1/\ell$ and tends to zero. Conversely, $\mu\ell \to \infty$ alone does not ensure $m \to 0$ if the prefactor μ grows sufficiently fast. The localization asymptotic controls the dimensionless suppression; an absolute mass limit needs a scaling hypothesis.

For the boundary-Yukawa problem, $X = 0$ gives exact zero modes of boundary types (a) and (b). With flavour-degenerate bulk masses, the boundary domain is invariant exactly when

$$V_Q X = X V_U.$$

If $N_Q = N_U$, this is the group of Proposition 17.5 for the corresponding limiting mass matrix. More generally the same formula defines the rectangular stabilizer.

The identification $M_F \propto X$ in lemma 12 is a **towerless-limit** statement, with $X = \ell\widehat{Y}$, and not an exact finite-ℓ mass formula. For one flavour and zero bulk masses, the exact light eigenvalue is $\arctan|X|/\ell$, not a constant multiple of $|X|$. The full boundary-coupled operator is a different construction from the fixed tensor-product truncation of Theorem 08.7; an identification of the two at finite ℓ requires additional proof.

Consequence for T10

Verdict: Error. The reduced action is a valid rewriting when the chosen admissible involution exists. The conclusion overstates its Dirac and KO-dimensional scope.

A general admissible reduction need not give one ordinary Dirac fermion. For example, take

$$\gamma_F = \sigma_3, \quad J_F = \sigma_1 K, \quad D_F = m\sigma_1, \quad U = \sigma_1.$$

Here $s = +1$, H_0 is one-dimensional, and the reduced action is

$$\tfrac{1}{2}\langle J_M\psi, (D_M + \gamma_5 m)\psi\rangle,$$

a Majorana-type Grassmann action with no second independent component from which to form the Dirac action of Corollary 17.2.

Moreover, the strict light spectrum in lemma 06, Result 7 has a 39-dimensional H_0. An antiunitary on an odd-dimensional complex space cannot square to -1, so that spectrum has no $s = -1$ reduction of the asserted type. Formal KO-label $4 + 5 = 9$ is available only when the necessary $s = -1$ involution exists; it does not establish a nine-dimensional metric geometry.

The $U(2) \times U(2)$ symmetry and $U(2)$ reduction space are correct for the colour-only, two-massless-flavour sector. The qualifications on alignment, spectral density and dynamics given above remain necessary.

Part 2(i): Coset and alignment

The massless coset has substantive equivariant content. At $M = 0$, the chiral group acts on reductions by

$$(V_L, V_R) : \Sigma \longmapsto V_L \Sigma V_R^\dagger.$$

The action is transitive, and the stabilizer of $\Sigma = 1$ is the diagonal $U(N_f)$. Thus the reductions form a homogeneous $U(N_f)_L \times U(N_f)_R$-space, not merely a set with the same dimension as a chiral coset.

The mass condition has an elementary linear-algebraic origin: it says that $H = \Sigma^\dagger M$ is Hermitian. For invertible M, $H^2 = M^\dagger M$, and the positive choice gives the polar unitary. The equality with the critical-point equation for $-\operatorname{Re}\operatorname{tr}(M\Sigma^\dagger)$ is exact. It identifies *all stationary points*, including saddles and maxima; vacuum alignment normally asks which stationary point minimizes a physical energy.

We therefore read the outcome paragraph as somewhat too strong when it says masses align the coset as a physical vacuum would. The remark correctly disclaims a proof of condensation, but identifying a reduction with a possible condensate direction still needs an explicit map to a quantum order parameter. The spectral action of the fixed even operator supplies no Σ-dependent potential merely from changing the reduction. In QCD, the anomalous singlet axial direction also means that the full classical $U(N_f)$ space cannot be identified without qualification with the pion manifold.

Part 2(ii): The sign, Pfaffian and anomaly

For the paired sector of Corollary 17.2, s is invisible in the classical fermion action **up to the displayed change of Grassmann variables**. The two reductions are coordinate descriptions of the same original bilinear form.

At finite dimension, if $B_- = W^T B_+ W$, then

$$\operatorname{Pf}(B_-) = \det(W)\operatorname{Pf}(B_+).$$

For the corollary one may take $W = \operatorname{diag}(1, \gamma_5)$. The Berezin measure transforms inversely, so a consistently transported measure gives the same integral. Separately chosen Pfaffian orientations or regulators can display a sign or phase; comparing those choices requires their specification. No physical anomaly or theta shift follows from changing the reduction sign s alone.

A **physical** change of a quark mass sign at fixed measure is different. For one massive fundamental Dirac flavour in a gauge background of index Q, the paired nonzero eigenvalues contribute equally for m and $-m$, while zero modes give the ratio

$$\frac{\det(\not{D}-m)}{\det(\not{D}+m)}=(-1)^{n_0}=(-1)^{Q}.$$

With the usual topological normalization, this is the factor associated with a π shift of θ. More generally, anomalous chiral rotations relate mass phases and θ; the conventionally invariant combination is written $\bar{\theta}=\theta+\arg\det M$, with signs depending on conventions. None of this makes s an additional observable.

Part 2(iii): Spacetime-dependent Σ

Let $c^\mu=[D_M,x^\mu]$ locally, and suppose $U(x)$ remains an admissible finite involution pointwise. Then

$$[D_M,\widetilde{U}]=c^\mu\otimes\partial_\mu U.$$

Set $B_\mu=(U\partial_\mu U)|_{H_0}$. With the same pointwise definition of R, direct differentiation gives

$$R^{-1}\widetilde{U}DR=D_M+c^\mu P_- B_\mu+\gamma_5 d(x).$$

Here $B_\mu^\dagger=-B_\mu$. Since $c^{\mu\dagger}=-c^\mu$ and $P_- c^\mu=c^\mu P_+$, the derivative term has adjoint $c^\mu P_+ B_\mu$. The displayed reduced operator is therefore generally **not self-adjoint**.

If $\Sigma(x)$ merely changes coordinates for the original fixed operator D, the original spectral action is independent of Σ. Likewise, conjugating D by a smooth unitary $U(x)$ preserves its spectrum exactly. Neither operation generates a genuine two-derivative kinetic term.

One can instead *define a new operator*

$$D_\Sigma^{\text{new}}=D_M+\gamma_5 d_\Sigma(x)$$

and omit the derivative term above. That is an additional choice, rather than the exact reduction of the fixed operator. For illustration, on a flat four-manifold, with $M=mI$, $\Sigma^\dagger=\Sigma$, $\Sigma^2=1$, and spectral action $\operatorname{Tr} f((D_\Sigma^{\text{new}})^2/\Lambda^2)$, its heat expansion contains

$$\frac{N_c f(0)m^2}{4\pi^2}\int \operatorname{tr}_f(\partial_\mu\Sigma^\dagger\partial^\mu\Sigma).$$

This normalization traces over both reduced particle and antiparticle blocks and uses $N_c=3$; its sign is positive when $f(0)>0$. At $M=0$, $d_\Sigma=0$, so this particular new operator has no Σ-dependence either. A pion kinetic term therefore requires a specified new operator or a quantum effective-action calculation.

Part 2(iv): Attributions

- The name **Banks–Casher** for the spectral-density relation is appropriate. We verified the bibliographic identity of Banks and Casher's 1980 paper, but its full text was inaccessible in this check; we do not attribute the precise hypotheses used above to that paper.
- **Vafa–Witten:** positivity of the Euclidean vectorlike fermion measure is part of the method described in Vafa and Witten, *Eigenvalue Inequalities for Fermions in Gauge Theories*, introduction and equation (1). Positivity alone should not be presented as the whole content of their results.
- **Dashen:** *Some Features of Chiral Symmetry Breaking*, Phys. Rev. D 3 (1971) 1879, especially the introduction and §II discusses the selection of a vacuum from a symmetry-limit family by an explicit perturbation. The matrix stationarity calculation above is our derivation, not an attribution of this reduction theorem to Dashen.
- **Kaplan and Shamir:** Kaplan's domain-wall construction uses chiral modes localized by a mass defect; Shamir's boundary construction uses boundaries of a finite or semi-infinite lattice. The interval calculation has the same localization and overlap mechanism, but is not literally either lattice operator.

Amendments and errors

- Require trace-class conditions in Theorem 17.1.
- State the change of independent Grassmann variables and measure conventions in Corollary 17.2.
- Describe JAJ^{-1} in Proposition 17.3 as having opposite-algebra **coefficients**.
- In Corollary 17.4, impose the residual gauge-group restriction, and replace the necessary-heavy-top claim by the weaker heavy-Dirac-mass conclusion under the matching hypotheses.
- In Theorem 17.6, retain $\Sigma^\dagger M = (\Sigma^\dagger M)^\dagger$ after imposing electric charge, and include the colour factors.
- Restrict Remark 17.7's negative-mass statement to invertible M, and distinguish a change of reduction from a physical mass-sign change.
- Add mass-limit, tail, regularization and limit-exchange hypotheses to Proposition 17.8; fix the determinant-positivity convention.
- Restrict Proposition 17.9's localization limit and its boundary mass identification to the regimes in which they are proved.
- Restrict the T10 Dirac and formal KO-9 conclusions to sectors that actually possess the required involution and paired fermion content.

Overall: 10 statements checked: 1 correct, 4 correct with amendment, 0 gaps, and 5 errors.